

# Revisiting Wage Rigidity and Welfare in Currency Unions: The Role of Downward Nominal Wage Rigidity\*

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## Abstract

This paper studies the welfare implications of downward nominal wage rigidity (DNWR) in a currency union. Using firm-level survey data from the European Union, we document the prevalence of DNWR and incorporate it into a small open economy New Keynesian model with staggered wage adjustment. Consistent with previous literature, a model without DNWR generates a hump-shaped relationship between welfare loss and wage sluggishness. Once DNWR is introduced and the constraint binds sufficiently frequently, however, this trade-off largely disappears. By constraining downward wage adjustment, DNWR compresses wage and price inflation volatility, making employment fluctuations the dominant source of welfare loss. As a result, welfare losses increase monotonically with wage sluggishness. We further show that these effects are state-dependent and that DNWR flattens the Phillips curve, amplifying the real effects of demand shocks in a currency union.

**Keywords:** Currency Union, European Union, Welfare, Wage Rigidity, Downward Wage Rigidity, Aggregate Demand and Supply

**JEL Classification:** E12, E24, E52, E63, F31, F33, F41

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# 1 Introduction

Nominal wage rigidity is often viewed as a source of economic inefficiency because it hinders labor-market adjustment and amplifies fluctuations in employment and output. However, [Galí and Monacelli \(2016\)](#) show that this conventional view need not hold in a currency union. In their framework, greater wage rigidity can improve welfare by stabilizing wage and price inflation, generating a hump-shaped relationship between welfare losses and wage rigidity. An important feature absent from this analysis, however, is downward nominal wage rigidity (DNWR), despite extensive evidence that wages are considerably more rigid downward than upward. This raises a natural question: How does DNWR affect the welfare implications of wage rigidity in a currency union?

This paper revisits the welfare implications of wage rigidity in a currency union by incorporating DNWR into the framework of [Galí and Monacelli \(2016\)](#). We first document that wage freezes are widespread across European countries and become particularly prevalent during downturns. Motivated by this evidence, we introduce DNWR into a small open economy New Keynesian (NK) model. Contrary to the findings of [Galí and Monacelli \(2016\)](#), we show that DNWR fundamentally changes the welfare implications of wage rigidity. In our baseline calibration, welfare losses increase monotonically with wage rigidity, implying that greater wage flexibility improves welfare. Importantly, welfare implications become state-dependent, depending on the frequency with which the DNWR constraint binds and therefore on the size, sign, and source of shocks.

We begin by documenting the prevalence of DNWR using data from the Wage Dynamics Network (WDN) survey. This firm-level survey, conducted across European Union (EU) countries, asks firms whether they have ever frozen or cut base wages for their employees. Using this data, we calculate the proportion of workers subjected to wage freezes year-on-year, as well as those experiencing wage cuts. Our analysis shows that in countries with a greater decline in the employment-to-population ratio, there was a corresponding increase in the share of workers with wage freezes relative to those with wage cuts, from 2007 to 2013, a period marked by rising unemployment rates across the EU—indicative of the negative aggregate shock following the Global Financial Crisis. These findings underscore the prevalence of DNWR, within the EU.

Motivated by our empirical findings, we develop a small open economy NK model with sticky price and wage setting that incorporates DNWR, building on the work of [Galí and Monacelli \(2016\)](#).

The model features two types of wage rigidity: households face both quadratic wage adjustment costs, proportional to the Calvo wage rigidity parameter, and DNWR. In equilibrium, downward wage adjustments are prohibited, while upward adjustments occur sluggishly.

Our main result is that DNWR changes the welfare implications of wage flexibility in a currency union. We compute welfare losses using a second-order approximation of the representative agent's utility and compare scenarios without and with DNWR. Consistent with [Galí and Monacelli \(2016\)](#), a model without DNWR generates a hump-shaped relationship between welfare loss and Calvo wage rigidity in a currency union, where monetary policy does not respond to domestic macroeconomic conditions. Greater wage flexibility reduces employment fluctuations but increases wage inflation volatility, creating a trade-off between quantity and price adjustment. Once DNWR is introduced, however, this trade-off largely disappears. By constraining downward wage adjustment, DNWR substantially compresses wage and price inflation volatility, making employment fluctuations the dominant source of welfare loss. As a result, welfare losses increase monotonically with Calvo wage rigidity, implying that greater wage flexibility is welfare-improving.

Our results highlight that the welfare effects of wage rigidity depend on the interaction between symmetric and asymmetric wage frictions. While greater Calvo wage rigidity increases welfare losses when DNWR is sufficiently relevant, introducing DNWR itself can reduce welfare losses by dampening wage and price inflation volatility. Thus, DNWR does not simply add another form of wage rigidity; it changes the relative importance of inflation and employment fluctuations in determining welfare outcomes.

Importantly, this result becomes state-dependent. The welfare implications of DNWR depend on the frequency with which the constraint binds, which is governed by many factors, including degree of DNWR, the size and sign of shocks, and the source of the shocks. When the DNWR constraint is frequently binding, inflation-related distortions become less important and employment fluctuations dominate welfare outcomes. When the constraint binds less often, the economy behaves more like the standard model without DNWR and the hump-shaped welfare trade-off re-emerges.

We further show that the monotonic welfare result is not a general property of DNWR itself, but arises from its interaction with monetary regimes in which stabilization policy is constrained. In a currency union, where nominal exchange-rate adjustment is absent, DNWR overturns the standard welfare trade-off when the DNWR constraint binds sufficiently frequently. A similar result emerges

at the zero lower bound (ZLB), where DNWR transforms the hump-shaped welfare loss function into a monotonic one.

Finally, we show that DNWR has important implications for macroeconomic stabilization. In a currency union, the aggregate demand curve is upward sloping, while DNWR flattens the aggregate supply (Phillips) curve. As a result, demand-side disturbances have larger effects on employment and output, highlighting how asymmetric wage rigidity alters macroeconomic stabilization even when monetary policy is unchanged.

**Related Literature and Contribution** Our paper contributes to several strands of literature. First, we provide cyclical evidence of DNWR in the European Union using micro-level data. [Dickens et al. \(2007\)](#) document asymmetries in wage change distributions across 16 countries, including European nations, noting a high incidence of wage freezes and a lack of nominal wage cuts, which is indicative of DNWR. [Schmitt-Grohé and Uribe \(2016\)](#) demonstrate that DNWR is pervasive in peripheral Europe, showing that aggregate nominal wages continued to grow in most peripheral European countries despite the large surge in unemployment during the Great Recession of 2008. Similarly, [Matschke \(2024\)](#) reports evidence of DNWR using aggregate wages adjusted for inflation across 53 countries, including those in Europe. Building on the work of [Branten et al. \(2018\)](#), who report the prevalence of DNWR by measuring the share of workers experiencing wage freezes relative to those receiving wage increases using WDN data, we extend their analysis by demonstrating how these shares are associated with employment across EU countries from 2007 to 2014. Our findings highlight the cyclical relationship between wage change distributions and employment, providing empirical evidence that DNWR becomes more relevant precisely when labor market conditions deteriorate.

Second, we contribute to the literature on the welfare implications of wage rigidity. [Galí and Monacelli \(2016\)](#) show that, in a currency union, welfare losses are hump-shaped in the degree of Calvo wage rigidity, while [Billi and Galí \(2020\)](#) obtain a similar trade-off in a closed economy at the zero lower bound. We show that introducing DNWR fundamentally alters this result. By compressing wage and price inflation volatility and shifting adjustment toward employment fluctuations, DNWR overturns the standard hump-shaped welfare trade-off and generates a monotonic welfare cost of wage rigidity under the baseline calibration. Importantly, we show that the welfare implications of

DNWR are state-dependent and governed by the frequency with which the constraint binds.

This paper also contributes to the literature on DNWR by emphasizing the role of the monetary policy regime in shaping its welfare implications. Prior work shows that the effects of DNWR are highly regime-dependent: while [Benigno and Ricci \(2011\)](#), [Mineyama \(2022\)](#), and [Aktug \(2025\)](#) highlight its role in generating benefits for positive inflation, [Amano and Gnocchi \(2023\)](#) and [Billi and Galí \(2020\)](#) show that its welfare effects depend critically on whether monetary policy is constrained, such as at the zero lower bound. We complement this literature by studying DNWR in a currency union and show that DNWR reshapes the welfare trade-off precisely because monetary policy cannot respond to country-specific conditions. While our paper investigates the role of DNWR in welfare within the context of monetary policy in a currency union, [Born et al. \(2024\)](#) examine the role of DNWR in a currency union from a fiscal policy perspective, documenting sign-dependent effects of government spending.

We also observe that the aggregate demand curve slopes upward in a currency union, with its slope minimally affected by the presence of DNWR. This phenomenon has also been noted in closed economies or small open economies during ZLB episodes ([Eggertsson, 2010](#); [Eggertsson and Krugman, 2012](#); [Eggertsson et al., 2014](#); [Eggertsson and Garga, 2019](#); [Bonciani and Oh, 2023a,b](#)). However, we show that this remains true in a small open economy within a currency union or under a pegged exchange rate regime, even outside of ZLB periods. Additionally, we find that DNWR flattens the Phillips curve in a currency union, a result that has been documented in closed economies ([Daly and Hobijn, 2014](#), [Mineyama, 2023](#)).

The remainder of the paper is organized as follows. Section 2 presents empirical evidence on wage rigidity across EU countries. Section 3 introduces the small open economy NK model with DNWR. Section 4 describes the calibration and solution methods. In Section 5, we present the results of our welfare analysis and discuss their policy implications. Section 6 examines how DNWR affects shock transmission through an aggregate demand–aggregate supply framework. Finally, Section 7 concludes the paper.

## 2 Empirical Evidence

Nominal wage rigidity, particularly downward nominal wage rigidity, is pervasive across European labor markets. Institutional features in several European countries make nominal wage reductions difficult, often requiring mutual consent between employers and workers.<sup>1</sup> To document the prevalence of wage rigidity across EU countries, we use data from the Wage Dynamics Network (WDN) surveys. The WDN is a collaborative research network that includes economists from the national central banks of EU member states and the European Central Bank (ECB). The surveys provide detailed information on firms' wage-setting practices.

The WDN conducted its surveys in three distinct waves-2007, 2009, and 2014-encompassing approximately 25,000 firms across 23 to 25 European countries. To ensure consistency and comparability among the participating nations, harmonized core questionnaires were developed collaboratively within the WDN framework. This standardized questionnaire was subsequently adopted by 13 of the participating countries.

Using WDN data, we construct country-level measures of the shares of workers experiencing wage freezes and wage cuts. The survey asks firms whether they froze or reduced base wages and, if so, the percentage of employees affected.<sup>2</sup> Using these responses, we first compute firm-level shares of workers affected by wage freezes and wage cuts and then aggregate them to the country level using employment weights.<sup>3</sup>

Note that the reference period differs in the 2007 survey wave. Firms were asked whether they had frozen or cut wages during the previous five years, whereas subsequent survey waves referred only to the previous year. Our results remain robust when the 2007 observations are excluded.

Despite being measured during an economic expansion, a non-negligible fraction of workers experienced no nominal wage change. On average, 6.5% of workers across countries were reported to have received no nominal wage adjustment during the five years preceding the 2007 survey. This

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<sup>1</sup>Austria (<https://www.deel.com/blog/how-to-reduce-payroll-costs-in-austria/>); Czech Republic (<https://cms.law/en/int/expert-guides/cms-expert-guide-to-labour-law-in-central-eastern-europe/czech-republic>); and France (<https://mgglegal.com/en/1576-2/>)

<sup>2</sup>The WDN survey focuses specifically on base wages. For instance, Wave 3 asked respondents: "Between 2010 and 2013, did you freeze or cut base wages in any year? If so, please indicate which year(s)." The question appears to refer to wage changes for incumbent employees, although the wording is somewhat ambiguous. Waves 1 and 2 contain similarly phrased questionnaires.

<sup>3</sup>Table A.1 presents these statistics for 13 countries from 2007 to 2013. Branten et al. (2018) report the ratio of these two statistics and interpret it as evidence of DNWR.

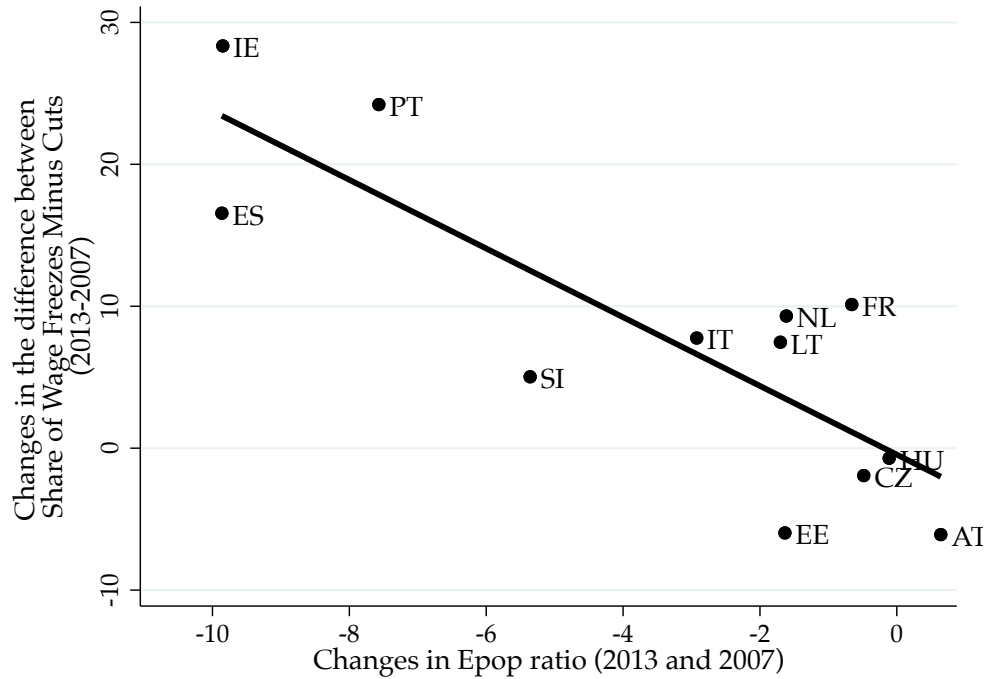


Figure 1: Change in Share of Wage Freezes and Cuts and Epop Ratio

Data Source: WEO and WDN. Notes: This graph depicts the change in the share of workers with wage freezes plotted against the change in the employment-to-population ratio across countries in the EU, with a fitted line. Each dot on the graph represents a country, with the respective abbreviations: AT for Austria, CZ for the Czech Republic, EE for Estonia, ES for Spain, FR for France, GR for Greece, HU for Hungary, IE for Ireland, IT for Italy, LT for Lithuania, NL for the Netherlands, PT for Portugal, and SI for Slovenia.

period corresponded to the expansionary phase preceding the global financial crisis, with the euro-area business cycle reaching its peak in 2008Q1.<sup>4</sup> The observation that workers experienced no nominal wage changes even during favorable economic conditions suggests the presence of persistent frictions in wage adjustment throughout the business cycle. Such behavior is consistent with time-dependent wage-setting models, such as Calvo wage rigidity, in which wages are adjusted only intermittently regardless of current economic conditions. Indeed, [Druant et al. \(2012\)](#) argue that wage adjustments in Europe exhibit a substantial degree of time dependence.

Additionally, to understand how nominal wage changes responded to the recession, we plot the change in the difference between the share of workers experiencing wage freezes and those experiencing wage cuts from 2007 to 2013, against changes in the employment-to-population (Epop) ratio over the same period (Figure 1). This period corresponds to the aftermath of the Global Financial

<sup>4</sup><https://eabcn.org/dbc/peaksandtroughs/chronology-euro-area-business-cycles>

Crisis, during which many EU countries faced sharp declines in employment rates—a clear manifestation of a negative aggregate shock. By focusing on the change in the difference between wage freezes and cuts, we effectively difference out country fixed effects. The vertical axis captures the extent to which wage freezes became more prevalent than wage cuts in response to deteriorating aggregate conditions during the 2007–2013 period.

If DNWR is a binding constraint, countries with a larger increase in wage freezes relative to wage cuts may have experienced greater allocative inefficiencies, resulting in a lower Epop ratio. Indeed, countries that saw larger declines in the employment-to-population ratio also exhibited more pronounced increases in the share of workers with wage freezes relative to cuts. For instance, between 2007 and 2013, Spain experienced a 10 percentage point larger drop in its employment-to-population ratio compared to Hungary. During the same period, the share of workers with wage freezes relative to cuts in Spain increased by 17 percentage points more than in Hungary.

To substantiate this finding for every year in the sample, we run the following regression equation:

$$\begin{aligned} [\text{Share of freezes}]_{it} &= \alpha_i + \gamma_t + \beta_s e_{it} + \delta \pi_{it} + \epsilon_{it}, \\ [\text{Share of wage cuts}]_{it} &= \alpha_i + \gamma_t + \beta_n e_{it} + \delta \pi_{it} + \epsilon_{it}, \end{aligned} \tag{1}$$

where  $\alpha_i$  and  $\gamma_t$  capture the country and time-fixed effects, respectively, and  $e_{it}$  represents the country-level employment-to-population ratio. The country-fixed effects account for country-specific characteristics, while the time-fixed effects capture factors common across countries in each year, including changes in survey reference periods and ECB monetary policy.

Table 1 reports how the distribution of nominal wage changes responds to the employment-to-population ratio. Controlling for country and time fixed effects, as well as inflation rates, we find that a 1 percentage point decrease in the Epop ratio is associated with a 2.9 percentage point increase in the share of workers with wage freezes, and a 0.8 percentage point increase in the proportion of workers experiencing wage cuts. These regression results suggest that countries experiencing a larger decline in the Epop ratio tend to see a greater rise in wage freezes compared to wage cuts—specifically, 2.1 percentage points more wage freezes than cuts for every 1 percentage point decline in employment.<sup>5</sup> Consistent with Jo (2024)—who shows that the excess sensitivity of wage

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<sup>5</sup>Our results remain robust when using the unemployment rate as a regressor, see Figure A.1 and Table A.2 in Appendix

Table 1: The Share of Wage Freezes and Cuts with Epop Ratio in EU

|                      | (1)<br>Share of Freezes<br>$\Delta W = 0$ | (2)<br>Share of Cuts<br>$\Delta W < 0$ | (3)<br>Share of Freezes Minus<br>Share of Cuts |
|----------------------|-------------------------------------------|----------------------------------------|------------------------------------------------|
| Epop Ratio (%)       | -2.857***<br>(0.599)                      | -0.763***<br>(0.148)                   | -2.085**<br>(0.777)                            |
| Inflation Rate       | 1.583<br>(1.457)                          | -0.696<br>(0.530)                      | 2.272<br>(1.895)                               |
| Country-Fixed Effect | Yes                                       | Yes                                    | Yes                                            |
| Time-Fixed Effect    | Yes                                       | Yes                                    | Yes                                            |
| Observations         | 72                                        | 71                                     | 71                                             |
| Adjusted $R^2$       | 0.61                                      | 0.83                                   | 0.50                                           |

Data Source: WEO, WDI, and WDN. Sample Period: 2007, 2009, 2010-2013. Sample Countries: Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Germany, Estonia, Spain, France, Greece, Croatia, Hungary, Ireland, Italy, Lithuania, Luxembourg, Latvia, Malta, Netherlands, Poland, Portugal, Romania, Slovenia, and United Kingdom. Country-level clustered robust standard errors are shown in parentheses.\*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\* $p < 0.01$ .

freezes relative to cuts is best explained by DNWR among the wage rigidity models—we interpret this pattern as evidence that DNWR becomes more relevant during downturns in EU countries. Motivated by these findings, we incorporate the DNWR constraint into the small open NK model in the following section.

Our empirical analysis documents rigidity in base wages; however, firms may still adjust total compensation through other channels. Indeed, [Babecký et al. \(2012\)](#) show that firms facing base wage rigidity are more likely to reduce bonuses or benefits, or slow promotions, as alternative means of labor cost adjustment. As a result, rigidity in total compensation is likely to be less pronounced than rigidity in base wages. Although our analysis does not account for changes in total compensation, the presence of base wage rigidity is nevertheless associated with allocative consequences for employment. This interpretation is consistent with [Grigsby et al. \(2021\)](#), who argue that changes in bonus payments function like temporary price discounts in the price rigidity literature and thus have limited impact on employment dynamics.

A.

### 3 Model

In this section, we introduce a DNWR constraint, following the approach of [Schmitt-Grohé and Uribe \(2016\)](#), into a small open economy NK model developed by [Galí and Monacelli \(2016\)](#). While [Galí and Monacelli \(2016\)](#) use the [Calvo \(1983\)](#) framework to introduce price and wage stickiness, we instead adopt the [Rotemberg \(1982\)](#) approach. Its symmetric equilibrium structure makes the model more tractable and facilitates the incorporation of DNWR. This model closely mirrors the framework proposed by [Amano and Gnocchi \(2023\)](#), who incorporate a DNWR constraint in a closed economy NK model.

#### 3.1 Firms

In the home economy, a continuum of firms, indexed by  $i \in [0, 1]$ , produces output. Each firm specializes in producing a differentiated good according to the production function:

$$Y_t(i) = A_t N_t(i)^{1-\alpha}, \quad (2)$$

where  $Y_t(i)$  represents the output of firm  $i$ , and  $N_t(i)$  is the labor input. The labor input  $N_t(i)$  is a constant elasticity of substitution aggregator of the different types of labor services  $N_t(i, j)$ , employed by firm  $i$  for each labor type  $j \in [0, 1]$ :

$$N_t(i) \equiv \left( \int_0^1 N_t(i, j)^{\frac{\epsilon_w - 1}{\epsilon_w}} dj \right)^{\frac{\epsilon_w}{\epsilon_w - 1}}, \quad (3)$$

where  $\epsilon_w > 1$  is the elasticity of substitution between the different types of labor services.  $A_t$  denotes the common stochastic productivity factor across all firms. The logarithm of productivity,  $a_t \equiv \log A_t$ , evolves according to an exogenous AR(1) process:

$$a_t = \rho_a a_{t-1} + \varepsilon_t^a, \quad (4)$$

where  $\rho_a$  represents the persistence of productivity shocks, and  $\varepsilon_t^a$  is an i.i.d. shock.

We can derive the labor demand function for each labor service type  $j$  from the firms' cost

minimization problem:

$$N_t(i, j) = \left( \frac{W_t(j)}{W_t} \right)^{-\epsilon_w} N_t(i), \quad (5)$$

for all  $i, j \in [0, 1]$ .  $W_t(j)$  denotes the nominal wage for labor type  $j$ . Consequently, the total demand for labor of type  $j$  across all firms is given by:

$$\mathcal{N}_t(j) = \int_0^1 N_t(i, j) di = \left( \frac{W_t(j)}{W_t} \right)^{-\epsilon_w} N_t, \quad (6)$$

where  $N_t \equiv \int_0^1 N_t(i) di$ . This labor demand equation serves as a constraint that type  $j$  workers face when setting their wages.

The present discounted value of current and future nominal profits for a firm is given by:

$$E_0 \left\{ \sum_{t=0}^{\infty} \Lambda_{0,t} \left( (1 + \tau) P_{H,t}(i) Y_t(i) - \int_0^1 W_t(j) N_t(i, j) dj - \frac{\phi_p}{2} \left( \frac{P_{H,t}(i)}{\Pi_H P_{H,t-1}(i)} - 1 \right)^2 P_{H,t} Y_t \right) \right\}, \quad (7)$$

where  $\Lambda_{0,t}$  is the stochastic discount factor for nominal payoffs and  $P_{H,t}(i)$  is the price of domestic variety  $i$ . Firms receive a production subsidy  $\tau$ . Firms incur non-negative, non-decreasing, and convex price adjustment costs based on firm  $i$ 's price inflation, measured in units of domestic currency. Firms set their nominal prices subject to a sequence of isoelastic demand schedules:

$$Y_t(i) = \left( \frac{P_{H,t}(i)}{P_{H,t}} \right)^{-\epsilon_p} Y_t, \quad (8)$$

where  $P_{H,t} \equiv \left( \int_0^1 P_{H,t}(i)^{1-\epsilon_p} di \right)^{\frac{1}{1-\epsilon_p}}$  is the domestic price index. In a symmetric equilibrium, the optimal pricing condition, i.e., the NK Phillips curve for price inflation, is as follows:

$$\begin{aligned} \phi_p \left( \frac{P_{H,t}}{\Pi_H P_{H,t-1}} - 1 \right) \frac{P_{H,t}}{\Pi_H P_{H,t-1}} &= \phi_p E_t \left\{ \Lambda_{t,t+1} \left( \frac{P_{H,t+1}}{\Pi_H P_{H,t}} - 1 \right) \left( \frac{P_{H,t+1}}{\Pi_H P_{H,t}} \right)^2 \frac{Y_{t+1}}{Y_t} \right\} \\ &\quad + \epsilon_p \frac{1}{\mathcal{M}_t^p} - (1 + \tau)(\epsilon_p - 1), \end{aligned} \quad (9)$$

where the price markup is given by  $\mathcal{M}_t^p \equiv (1 - \alpha) \frac{P_{H,t}}{W_t} \frac{A_t}{N_t^\alpha}$ .

### 3.2 Households

The household is composed of a continuum of members, indexed by  $j \in [0, 1]$ . Each household member is specialized in a differentiated occupation and supplies labor services in an amount  $\mathcal{N}_t(j)$ .

Household preferences are represented by the following utility function:

$$E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, \{\mathcal{N}_t(j)\}; Z_t), \quad (10)$$

where  $C_t$  is a consumption index and  $Z_t$  is an exogenous preference shifter.  $\beta$  is the discount factor.

The period utility function  $U$  is specified as:

$$U(C_t, \{\mathcal{N}_t(j)\}; Z_t) = \left( \log C_t - \frac{1}{1+\varphi} \int_0^1 \mathcal{N}_t(j)^{1+\varphi} dj \right) Z_t, \quad (11)$$

with the consumption index  $C_t$  is given by:

$$C_t \equiv \frac{C_{H,t}^{1-v} C_{F,t}^v}{(1-v)^{1-v} v^v}, \quad (12)$$

Here,  $C_{H,t}$  is an index of domestic goods consumption, defined as  $C_{H,t} \equiv \left( \int_0^1 C_{H,t}(i)^{\frac{\epsilon_p-1}{\epsilon_p}} di \right)^{\frac{\epsilon_p}{\epsilon_p-1}}$ , where  $i \in [0, 1]$  represents the variety of domestic goods, and  $C_{F,t}$  denotes the quantity consumed of a composite foreign good. The parameter  $\epsilon_p > 1$  denotes the elasticity of substitution between domestically produced varieties, while  $v \in [0, 1]$  is a measure of openness to foreign goods.

The log preference shifter,  $z_t \equiv \log Z_t$ , is assumed to follow an exogenous AR(1) process:  $z_t = \rho_z z_{t-1} + \varepsilon_t^z$ . Shocks to  $z_t$  affect the marginal rate of substitution of consumption across time, thereby generating fluctuations in aggregate demand for a given prevailing interest rate. Accordingly, we refer to them as demand shocks.

The optimal allocation of any given expenditure on domestic goods yields the demand functions:

$$C_{H,t}(i) = \left( \frac{P_{H,t}(i)}{P_{H,t}} \right)^{-\epsilon_p} C_{H,t}, \quad (13)$$

for all  $i \in [0, 1]$ . Thus, we obtain  $\int_0^1 P_{H,t}(i) C_{H,t}(i) di = P_{H,t} C_{H,t}$ . The optimal allocation of expen-

ditures between domestic and imported goods is given by:

$$P_{H,t}C_{H,t} = (1 - v) P_t C_t, \quad (14)$$

$$P_{F,t}C_{F,t} = v P_t C_t, \quad (15)$$

where  $P_t \equiv P_{H,t}^{1-v} P_{F,t}^v$  is the consumer price index. Note that parameter  $v$  corresponds to the share of domestic consumption allocated to imported goods and can thus be interpreted as an index of openness.

The household's period budget constraint is as follows:

$$\begin{aligned} \int_0^1 P_{H,t}(i) C_{H,t}(i) di + P_{F,t}C_{F,t} + W_t \int_0^1 \frac{\phi_w}{2} \left( \frac{W_t(j)}{\Pi^w W_{t-1}(j)} - 1 \right)^2 dj + E_t \{ \Lambda_{t,t+1} D_{t+1} \} \\ \leq D_t + \int_0^1 W_t(j) \mathcal{N}_t(j) dj - T_t, \end{aligned} \quad (16)$$

where  $P_{F,t}$  represents the price of the imported good and  $D_{t+1}$  is the nominal payoff in period  $t + 1$  of the portfolio held at the end of period  $t$ , which may include shares in domestic firms. Households incur quadratic wage adjustment costs, modeled as a non-negative, non-decreasing, and convex function  $\frac{\phi_w}{2} \left( \frac{W_t(j)}{\Pi^w W_{t-1}(j)} - 1 \right)^2$  of household  $j$ 's wage inflation.  $T_t$  refers to lump-sum taxes. All previous variables are expressed in units of domestic currency. The stochastic discount factor for one-period ahead nominal payoff is given by  $\Lambda_{t,t+1} \equiv \beta \frac{C_t}{C_{t+1}} \frac{P_t}{P_{t+1}}$ .

Additionally, we assume that nominal wages are subject to a DNWR constraint, as suggested by [Schmitt-Grohé and Uribe \(2016\)](#):

$$W_t(j) \geq \gamma W_{t-1}(j), \quad (17)$$

where the parameter  $\gamma$  governs the degree of DNWR. There are two types of wage rigidity in the model: quadratic wage adjustment cost and DNWR. In equilibrium, households pay quadratic adjustment costs for upward wage adjustments, while downward wage adjustments are not allowed, given  $\gamma = 1$ .

We assume that the law of one price holds, implying that the price of imported goods, expressed in domestic currency, is given by:  $P_{F,t} = \mathcal{E}_t P_t^*$ , where  $\mathcal{E}_t$  is the nominal exchange rate (the price of foreign currency in terms of domestic currency) and  $P_t^*$  is the foreign price level (in foreign

currency). We assume that the foreign price level remains constant and normalized to one, i.e.,  $P_t^* = 1$ , for all  $t$ .

The household's intertemporal optimality condition takes the form:

$$1 = \beta (1 + i_t) E_t \left\{ \frac{C_t}{C_{t+1}} \frac{Z_{t+1}}{Z_t} \frac{P_t}{P_{t+1}} \right\}, \quad (18)$$

where  $i_t$  is the interest rate on a one-period nominal risk-free bond denominated in domestic currency. A similar condition holds with regard to bonds denominated in foreign currency, suitably adjusted for exchange rate movements:

$$1 = \beta (1 + i_t^*) E_t \left\{ \frac{C_t}{C_{t+1}} \frac{Z_{t+1}}{Z_t} \frac{P_t}{P_{t+1}} \frac{\mathcal{E}_{t+1}}{\mathcal{E}_t} \right\}. \quad (19)$$

By combining the above two conditions, we obtain:

$$1 = \beta E_t \left\{ \frac{C_t}{C_{t+1}} \frac{Z_{t+1}}{Z_t} \frac{P_t}{P_{t+1}} \left( 1 + i_t - (1 + i_t^*) \frac{\mathcal{E}_{t+1}}{\mathcal{E}_t} \right) \right\}. \quad (20)$$

We assume there is a complete set of state-contingent securities traded internationally. Utility maximization by the domestic households implies the following condition:

$$\frac{\mathcal{E}_t V_{t,t+1}}{P_t} \frac{Z_t}{C_t} = \xi_{t,t+1} \beta \frac{Z_{t+1}}{C_{t+1}} \frac{\mathcal{E}_{t+1}}{P_{t+1}}, \quad (21)$$

where  $V_{t,t+1}$  is the period  $t$  price (in foreign currency) of a one-period Arrow security that pays one unit of foreign currency if a specific state of nature is realized in period  $t+1$ , and nothing otherwise.  $\xi_{t,t+1}$  represents the probability of that state of nature being realized in  $t+1$ , conditional on the state of nature at  $t$ . A similar condition holds for foreign households:

$$\frac{V_{t,t+1}}{P_t^*} \frac{Z_t^*}{C_t^*} = \xi_{t,t+1} \beta \frac{Z_{t+1}^*}{C_{t+1}^*} \frac{1}{P_{t+1}^*}. \quad (22)$$

$C_t^*$  is (per capita) world consumption, and  $Z_t^*$  is a discount factor shock in the rest of the world. By

combining these two conditions, we obtain a risk sharing condition of the form:

$$C_t = \vartheta C_t^* Q_t \frac{Z_t}{Z_t^*}, \quad (23)$$

where  $Q_t \equiv \frac{\varepsilon_t P_t^*}{P_t}$  is the real exchange rate. Without loss of generality we set  $\vartheta = 1$ .

Each household member specialized in each occupation (or a union representing them) sets the corresponding nominal wage, subject to DNWR (17) and an isoelastic demand function for their services (6). In a symmetric equilibrium, the optimal wage-setting condition, i.e., the NK Phillips curve for wage inflation, is as follows:

$$\begin{aligned} \phi_w \left( \frac{W_t}{\Pi^w W_{t-1}} - 1 \right) \frac{W_t}{\Pi^w W_{t-1}} - \Delta_t = E_t \left\{ \Lambda_{t,t+1} \left( \phi_w \left( \frac{W_{t+1}}{\Pi^w W_t} - 1 \right) \left( \frac{W_{t+1}}{\Pi^w W_t} \right)^2 - \gamma \Delta_{t+1} \right) \right\} \\ + \left( \epsilon_w \frac{1}{\mathcal{M}_t^w} - \epsilon_w + 1 \right) N_t, \end{aligned} \quad (24)$$

where wage markup is defined by  $\mathcal{M}_t^w \equiv \frac{W_t}{P_t} \frac{1}{N_t^\varphi C_t}$ .  $\Delta_t$  denotes the Lagrange multiplier associated with the DNWR constraint (17). It captures the marginal benefit of relaxing the constraint and satisfies the standard complementary slackness and nonnegativity conditions:

$$\Delta_t (W_t - \gamma W_{t-1}) = 0, \quad \Delta_t \geq 0, \quad W_t - \gamma W_{t-1} \geq 0. \quad (25)$$

### 3.3 Exports Demand

Prices are assumed to be the same for both domestic and export markets, implying that the law of one price holds for exports as well. The demand for exports of domestic good  $i$  is given by:

$$X_t(i) = \left( \frac{P_{H,t}(i)}{P_{H,t}} \right)^{-\epsilon_p} X_t, \quad (26)$$

for  $i \in [0, 1]$ , where  $X_t$  is an aggregate export index. The aggregate export index is assumed to be given by:

$$X_t = v S_t Y_t^*, \quad (27)$$

where  $Y_t^*$  is world output (expressed in per capita terms) and  $S_t \equiv \frac{P_{F,t}}{P_{H,t}}$  denotes the terms of trade. In equilibrium, world output,  $Y_t^*$ , equals world consumption,  $C_t^*$ . We also consider a symmetric

steady state where  $S = 1$  and  $C = C^* = Y^*$ . In this case,  $X = vY^*$  and  $C_F = vC$ , implying a balanced trade, as well as  $Y = C$ , in the steady state.

### 3.4 Global Shocks

We consider two types of global shocks that impact the home economy, referred to as export shocks and world interest rate shocks. Positive export shocks raise demand for exports of domestic goods, shown in Equation (27), leaving the world interest rate unchanged, serving as an aggregate demand shock for home economy. On the contrary, world interest rate shocks change the world real interest rate, leaving global output unchanged (thus influencing exports only through an eventual endogenous response of the terms of trade). While world output and the world interest rate are endogenous and likely correlated, we aim to isolate their respective effects on the home economy. To do so, we introduce global shocks into the model as follows. We assume that the discount factor shifter for foreign households is given by:

$$Z_t^* = Z_{1,t}^* Z_{2,t}^*, \quad (28)$$

where  $z_{1,t}^* \equiv \log Z_{1,t}^*$  and  $z_{2,t}^* \equiv \log Z_{2,t}^*$  follow independent, exogenous AR(1) processes:  $z_{i,t}^* = \rho_i^* z_{i,t-1}^* + \varepsilon_{i,t}^*$ , for  $i = 1, 2$ . Exogenous shocks  $Z_{1,t}^*$  and  $Z_{2,t}^*$  are characterized by the differing monetary policy responses they elicit from the foreign central bank. Specifically, we assume that the world real interest rate remains unchanged in response to  $Z_{1,t}^*$  shocks. In contrast, the foreign central bank adjusts the real interest rate in response to  $Z_{2,t}^*$  shocks to maintain a constant global output level. Assuming that foreign households have an Euler equation similar to domestic households:

$$1 = \beta (1 + i_t^*) E_t \left\{ \frac{Y_t^*}{Y_{t+1}^*} \frac{Z_{t+1}^*}{Z_t^*} \right\}, \quad (29)$$

it follows, along with the global market clearing condition  $C_t^* = Y_t^*$ , that:  $Y_t^* = Z_{1,t}^*$ . This implies that the risk-sharing condition can be expressed as:

$$C_t = Q_t \frac{Z_t}{Z_{2,t}^*}. \quad (30)$$

The world interest rate, according to the assumptions above, is given by:

$$i_t^* = \rho + (1 - \rho_2^*) z_{2,t}^*. \quad (31)$$

Shocks to  $z_{1,t}^*$  directly influence global output, thereby affecting the demand for home exports as described in Equation (27). On the other hand, shocks to  $z_{2,t}^*$  alter the world real interest rate and shift the risk-sharing condition (30). These shocks impact aggregate exports indirectly, primarily through their effect on the real exchange rate. This distinction supports our classification of these shocks as export shocks and world interest rate shocks, respectively.

### 3.5 Monetary Regime

In a currency union, the home economy is assumed to be part of a global currency union, or it credibly pegs its exchange rate to the world currency. In either case, we let  $e_t \equiv \log \mathcal{E}_t$  denote the log nominal exchange rate, and assume, without loss of generality, that:

$$e_t = 0, \quad (32)$$

for all  $t$ . Under this regime, the domestic nominal interest rate moves one-for-one with the world interest rate, irrespective of domestic economic conditions.

To explore the role of monetary policy in our framework, we will later introduce an alternative regime: inflation targeting. In this regime, the central bank aims to stabilize domestic inflation  $\pi_{H,t} \equiv p_{H,t} - p_{H,t-1}$ , while allowing the exchange rate to float freely. Specifically, we assume

$$\pi_{H,t} = 0, \quad (33)$$

for all  $t$ . While these two regimes represent extreme scenarios, they simplify the analysis and help illustrate our key points more clearly.

### 3.6 Market Clearing

We focus on a symmetric equilibrium, for all  $t$ ,  $i$ , and  $j$ . Goods market clearing in the home economy requires the following conditions to hold:

$$Y_t = A_t N_t^{1-\alpha}, \quad (34)$$

$$\begin{aligned} Y_t &= C_{H,t} + X_t + \frac{W_t}{P_{H,t}} \frac{\phi_w}{2} \left( \frac{W_t}{\Pi^w W_{t-1}} - 1 \right)^2 + Y_t \frac{\phi_p}{2} \left( \frac{P_{H,t}}{\Pi_H P_{H,t-1}} - 1 \right)^2 \\ &= (1-v) \mathcal{S}_t^v C_t + v \mathcal{S}_t Y_t^* + \frac{W_t}{P_{H,t}} \frac{\phi_w}{2} \left( \frac{W_t}{\Pi^w W_{t-1}} - 1 \right)^2 + Y_t \frac{\phi_p}{2} \left( \frac{P_{H,t}}{\Pi_H P_{H,t-1}} - 1 \right)^2. \end{aligned} \quad (35)$$

The linearized equilibrium conditions that describe the equilibrium of the small open economy are detailed in Appendix D.

### 3.7 Welfare Loss Function

Under the assumption of an efficient steady state  $\left( \tau = \left( \frac{\epsilon_p}{\epsilon_p - 1} \right) \left( \frac{\epsilon_w}{\epsilon_w - 1} \right) (1-v) - 1 \right)$  and the zero inflation steady state, the average period utility losses of the small open economy's representative household can be derived following [Woodford \(2003\)](#) and [Galí and Monacelli \(2016\)](#).

$$\mathbb{W}_0 \equiv E_0 \sum_{t=0}^{\infty} \beta^t U(C_t, \{N_t(j); Z_t\}) = -\frac{1-v}{2} E_0 \sum_{t=0}^{\infty} \beta^t \mathbb{L}_t + t.i.p. + O(\|\xi\|^3), \quad (36)$$

The second-order approximated welfare loss function is as follows:

$$\mathbb{L}_t \approx (1+\varphi) \tilde{n}_t^2 + \frac{\epsilon_p}{\lambda_p (1-\alpha)} \pi_{H,t}^2 + \frac{\epsilon_w}{\lambda_w} \pi_t^{w2}, \quad (37)$$

where  $\lambda_p \equiv \frac{\epsilon_p Y}{\phi_p} = \frac{(1-\theta_p)(1-\theta_p\beta)(1-\alpha)}{\theta_p(1-\alpha+\alpha\epsilon_p)}$  and  $\lambda_w \equiv \frac{\epsilon_w N}{\phi_w} = \frac{(1-\theta_w)(1-\theta_w\beta)}{\theta_w(1+\epsilon_w\varphi)}$ .

The second-order average welfare loss function is a linear combination of the variances of the employment gap ( $\tilde{n}_t \equiv n_t - n_t^n$ , with  $n_t^n$  being the natural level of employment), domestic price inflation ( $\pi_{H,t}^p$ ), and wage inflation ( $\pi_t^w$ ).<sup>6</sup> The presence of a DNWR constraint does not directly affect either the functional form of the second-order welfare approximation or the coefficients on the variance terms ([Amano and Gnocchi, 2023](#)). However, the simulated path used to evaluate the

<sup>6</sup>Lowercase letters denote the natural logarithms of the original variables and with constants ignored.

welfare loss function—obtained from the piecewise linear solution of the model—does incorporate the DNWR constraint. Accordingly, we evaluate the second-order welfare approximation along the piecewise-linear solution path. This approach abstracts from higher-order effects, such as precautionary wage-setting behavior under DNWR. This method of computing welfare loss is widely used in NK models with occasionally binding constraints, particularly in analyses of the welfare implications of the zero lower bound (ZLB), for example, [Eggertsson and Woodford \(2003\)](#), [Nakata et al. \(2019\)](#), [Billi and Galí \(2020\)](#), and [Bonciani and Oh \(2025\)](#).

Although we incorporate Rotemberg-style wage adjustment costs, interpreting the parameter  $\phi_w$  structurally is challenging, as it lacks a direct empirical counterpart. In contrast, the Calvo wage rigidity parameter  $\theta_w$  maps directly to the expected duration of wage spells. Additionally, to facilitate comparison with earlier studies based on Calvo wage rigidity, we interpret  $\phi_w$  in terms of its Calvo counterpart  $\theta_w$ . Accordingly, we present the welfare loss function (37) in terms of the Calvo rigidity parameters  $\theta_p$  and  $\theta_w$ , as well as the Rotemberg adjustment parameters  $\phi_p$  and  $\phi_w$ , using first-order equivalences between the Calvo and Rotemberg frameworks (see, e.g., [Rotemberg, 1987](#) and [Roberts, 1995](#)). This substitution preserves the model’s welfare implications: [Nisticó \(2007\)](#) and [Oh \(2025\)](#) show that, under efficient steady-state conditions with revenue subsidies to firms, the two sticky price models yield identical inflation–output dynamics and welfare loss functions.

## 4 Calibration and Solution Method

Table 2 reports the calibrated parameters. To ensure a fair comparison, we closely adhere to the parameter calibration outlined by [Galí and Monacelli \(2016\)](#).

We introduce a new parameter  $\gamma$ , which governs the degree of DNWR and is calibrated to  $\gamma = 1$ . While the WDN data provide micro-data evidence of DNWR, mapping this evidence directly into the model parameter is not straightforward. First, the WDN surveys report the incidence of wage freezes and wage cuts, but not the magnitude of wage adjustments. As a result, they do not provide sufficient information to recover the full distribution of wage changes required for a direct calibration of  $\gamma$ . Second, our model is formulated in a representative-agent framework, whereas the WDN evidence reflects cross-sectional heterogeneity across firms and workers. Establishing a structural mapping between the two would therefore require a heterogeneous-agent environment

with occasionally binding constraints, which is beyond the scope of this paper.

To provide empirical discipline, we instead follow [Schmitt-Grohé and Uribe \(2016\)](#) and use aggregate nominal wage growth during downturns to calibrate  $\gamma$ , guided by the model. In the model, wage growth equals  $\gamma$  when the DNWR constraint binds, that is, when its multiplier is positive ( $\Delta_t > 0$  and  $\gamma = W_t/W_{t-1}$  in Equation (17)). Because these are states in which the marginal benefit of relaxing the constraint is positive, we associate them with periods of rising unemployment and proxy  $\gamma$  with observed wage growth in those periods. When the constraint is slack, wage growth satisfies  $W_t > \gamma W_{t-1}$  and is uninformative about  $\gamma$ ; including such periods would bias the implied values upward. Table A.3 reports the implied quarterly values of  $\gamma$  across countries in our sample, netting out long-run TFP growth.<sup>7</sup> On average, the implied  $\gamma$  is very close to one, indicating that aggregate nominal wages tend not to decline even during downturns. These estimates lie within the range of [0.990, 1.022] that [Schmitt-Grohé and Uribe \(2016\)](#) document for the euro-area periphery over 2008:Q1–2011:Q2.<sup>8</sup> We treat these statistics as descriptive discipline for the calibration of  $\gamma$  rather than as a point estimate; aggregate wage growth can reflect heterogeneity, composition effects, and productivity trends.

Based on this evidence, we set  $\gamma = 1$  in the baseline specification. However, we acknowledge that aggregate wage growth may overstate the degree of DNWR because composition effects can raise average wages during downturns as lower-wage workers disproportionately exit employment. For this reason, we complement the aggregate-based calibration with a more conservative robustness exercise using  $\gamma = 0.99$ , the value implied for Greece during the Great Recession, which [Schmitt-Grohé and Uribe \(2016\)](#) refer to as “the lowest of the cross-country estimates.”

In our sample, Greece has the lowest implied values, at slightly above 0.99. This is consistent with the WDN evidence, which shows that Greece experienced the highest incidence of wage cuts (29% in 2012 and 21% in 2013 as shown in Table A.1), while other countries report wage cuts affecting less than 5% of workers. Greece’s unemployment rate reached 27.7% in 2013. Our main results remain robust under this alternative calibration once shocks are large enough for the constraint to

<sup>7</sup>Our model abstracts from long-run productivity growth, which can raise nominal wages in the data; we therefore subtract each country’s average TFP growth from observed wage growth, motivated by [Schmitt-Grohé and Uribe \(2016\)](#).

<sup>8</sup>As an alternative, we condition on a high reported incidence of wage freezes rather than on unemployment: in WDN country-years in which more than 20% of workers were subject to a base-wage freeze (the 75th percentile of the country-year distribution), median annual wage growth is approximately 0.14%, corresponding to an implied quarterly  $\gamma$  of 1.0003. Among country-years with freeze shares of at least 40% (roughly the 90th percentile of the distribution), the median is approximately  $-0.37\%$ , corresponding to 0.9991.

Table 2: Quarterly Calibration

| Parameters     | Description                          | Value           |
|----------------|--------------------------------------|-----------------|
| $\gamma$       | Downward nominal wage rigidity       | 1               |
| $\varphi$      | Curvature of labor disutility        | 2.2             |
| $\alpha$       | Index of decreasing returns to labor | 0.26            |
| $\epsilon_p$   | Elasticity of substitution (goods)   | 3.8             |
| $\epsilon_w$   | Elasticity of substitution (labor)   | 4.3             |
| $\theta_p$     | Degree of Calvo price rigidities     | 0.8             |
| $\phi_p$       | Degree of Rotemberg price rigidities | 170.64          |
| $\theta_w$     | Degree of Calvo wage rigidities      | [0.1, 0.9]      |
| $\phi_w$       | Degree of Rotemberg wage rigidities  | [5.55, 3713.78] |
| $\Pi_H, \Pi^w$ | Steady-state inflation               | 1               |
| $v$            | Openness                             | 0.3             |
| $\beta$        | Discount factor                      | 0.99            |
| $\rho_i$       | Persistence of exogenous processes   | 0.9             |
| $\sigma_i$     | Volatility of exogenous processes    | 0.01            |

bind sufficiently frequently.

We solve for the occasionally binding DNWR constraint using the OccBin toolkit developed by [Guerrieri and Iacoviello \(2015\)](#) in the Dynare software package ([Adjemian et al., 2026](#)), which delivers a piecewise-linear solution. This approach omits the precautionary wage-setting behavior that a fully global method would capture—namely, wage setters’ restraint in raising wages today in anticipation that the DNWR constraint may bind in the future.

## 5 Welfare Implications

To assess the role of DNWR in explaining welfare losses, we compare the welfare losses for a small open economy in a currency union without and with the DNWR constraint. For each parameter setting, we simulate equilibrium paths over 10,000 periods and compute the variances of the employment gap, price inflation, and wage inflation using the simulated data. Welfare losses are then calculated for each configuration using Equation (37).

### 5.1 Benchmark: Without DNWR

Figure 2 shows the average welfare loss as a function of Calvo wage rigidity parameter,  $\theta_w$ , without the DNWR constraint. Each panel presents welfare loss functions conditional on one of the four

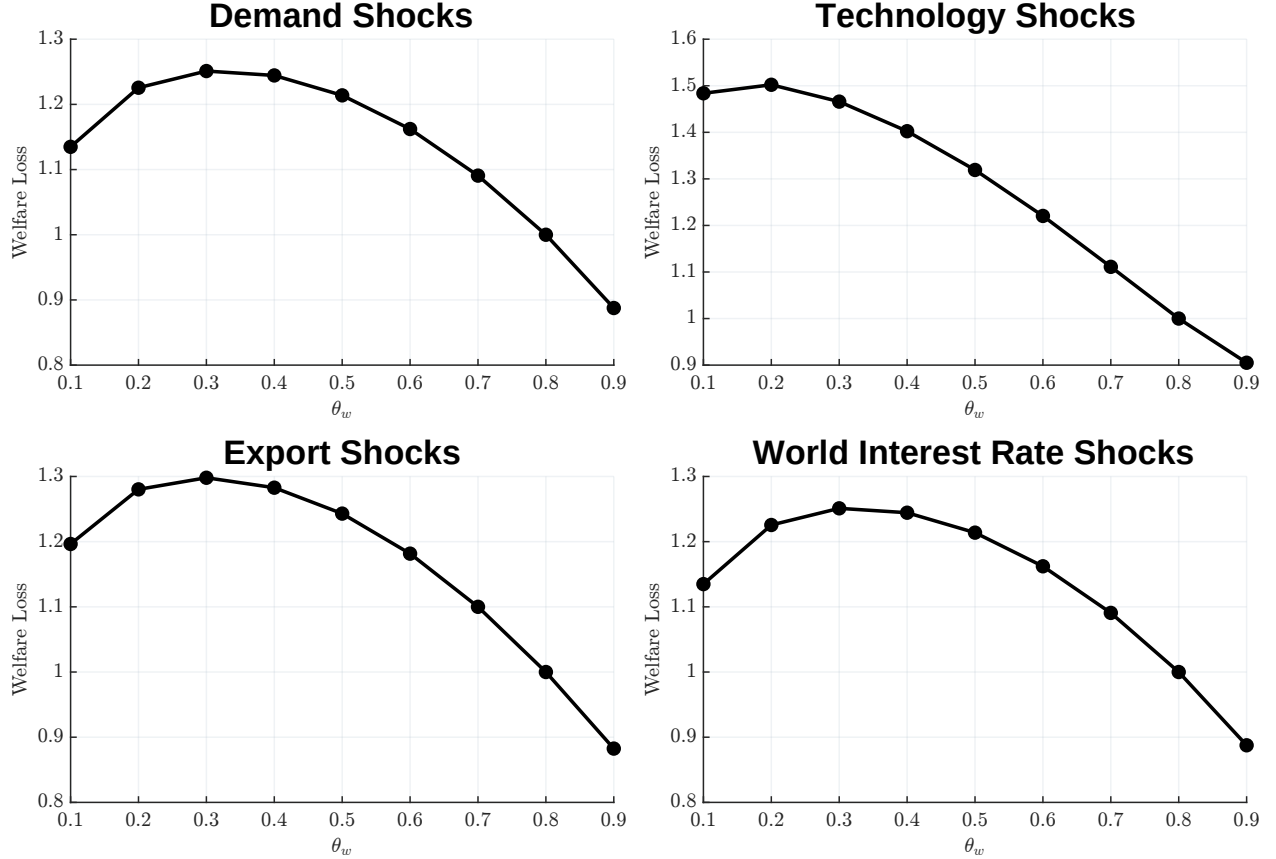


Figure 2: Welfare Loss across Different Shocks with Varying Calvo Wage Rigidity Parameters without DNWR

Note: This graph plots the welfare loss function based on the model without DNWR, against the Calvo wage rigidity parameter ( $\theta_w$ ) across different types of shocks: demand (top-left), technology (top-right), export (bottom left), and world interest rate shocks (bottom right). In each case, the welfare losses are expressed as a ratio relative to those under the baseline setting  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), while all other parameters are fixed at baseline values. These results are aligned with Figure 3 in [Galí and Monacelli \(2016\)](#).

exogenous shocks: demand shocks (top-left), technology shocks (top-right), export shocks (bottom-left), and world interest rate shocks (bottom-right). All parameters, except for  $\theta_w$ , are set to their baseline values. In each case, the welfare losses are expressed as a ratio relative to those under the baseline setting  $\theta_w = 0.8$ . Accordingly, our Figure 2 corresponds to Figure 3 in [Galí and Monacelli \(2016\)](#). The one difference is that we adopt Rotemberg wage adjustment costs rather than Calvo wage rigidity. Although the model is implemented using Rotemberg adjustment costs for tractability, we interpret the results on the scale of Calvo wage rigidity to ensure comparability with [Galí and Monacelli \(2016\)](#) and to provide meaningful interpretation of parameter values. For example,  $\theta_w = 0.8$  implies that wage setters are unable to adjust their wages 80% of the time each quarter, while

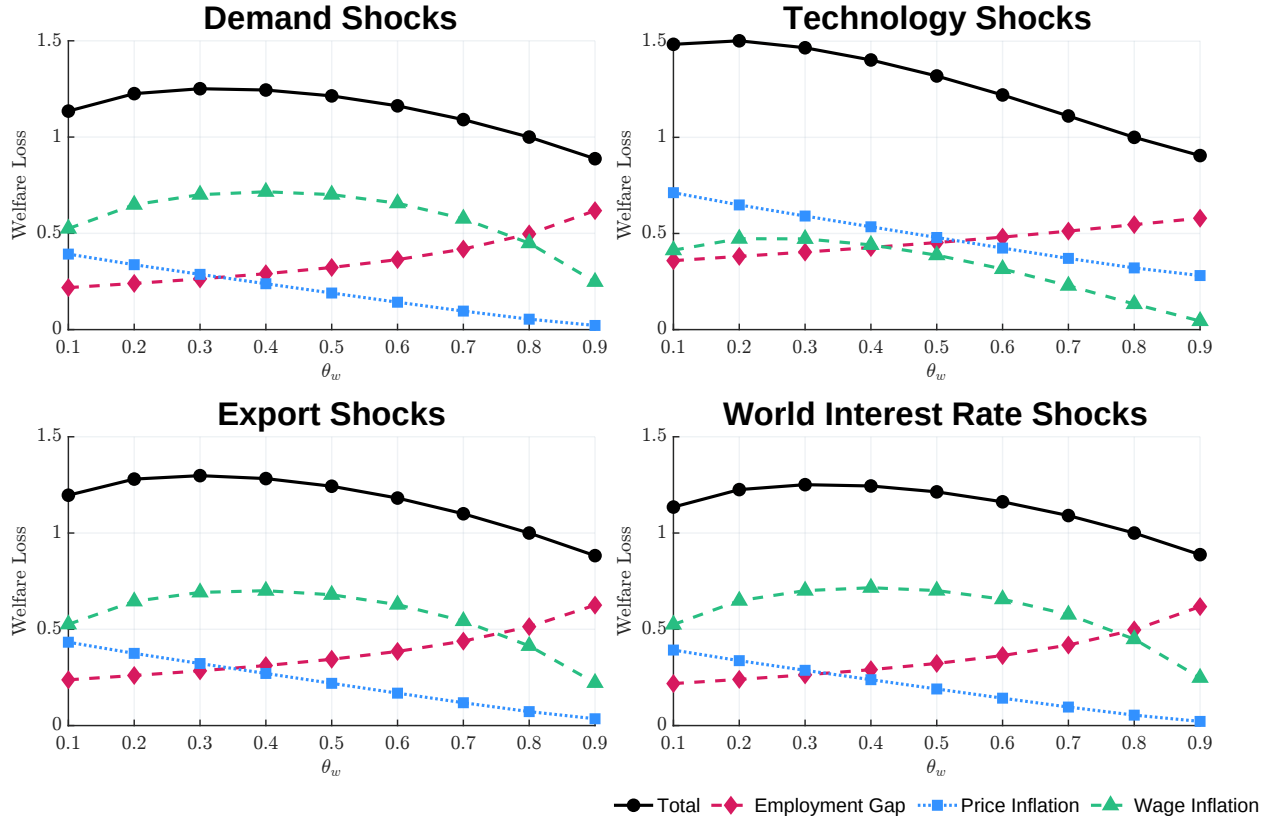


Figure 3: Welfare Loss Decomposition across Different Shocks with Varying Calvo Wage Rigidity Parameters without DNWR

Note: This figure illustrates the decomposition of the welfare loss function based on the model without DNWR against the Calvo wage rigidity parameter ( $\theta_w$ ) across different types of shocks: demand (top-left), technology (top-right), export (bottom left), and world interest rate shocks (bottom right). Each line represents the welfare loss attributed to the employment gap (red dashed line with diamonds), price inflation (blue dotted line with squares), and wage inflation (green dashed line with triangles). For each shock, total welfare loss is normalized by the baseline value corresponding to  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), while holding all other parameters fixed. Each component of welfare loss is also normalized by the baseline total welfare loss so that the sum of the components equals total welfare loss for each shock.

the corresponding value of  $\phi_w = 864.96$  is difficult to interpret on its own. A comparison of the two figures confirms that this modeling substitution does not alter the welfare results.<sup>9</sup>

Consistent with their findings, the total welfare loss function exhibits a hump-shaped curve, peaking at  $\theta_w$  values of 0.3, 0.2, 0.3, and 0.3 conditioning on demand, technology, export, and world interest rate shocks, respectively. This finding leads Galí and Monacelli (2016) to conclude

<sup>9</sup>Our Figure 2 successfully replicates Figure 3 from Galí and Monacelli (2016), except for the export shocks. Specifically, Panel C of Figure 3 in Galí and Monacelli (2016) depicts the welfare loss function as monotonically increasing with respect to the Calvo wage rigidity parameter in response to export shocks. However, our figure reveals a hump-shaped welfare loss function in response to export shocks. This discrepancy arises from a minor coding error in Galí and Monacelli (2016), which the authors have kindly acknowledged. Upon correcting this error, their finding of a hump-shaped welfare loss function becomes even more robust, applying to all four shocks. This reinforces their argument that higher wage rigidity can enhance welfare regardless of the source of the shock.

that increasing wage flexibility (i.e. lowering  $\theta_w$ ) can raise welfare loss as soon as  $\theta_w$  falls below a value close to 1.

To understand the sources of the hump-shaped welfare loss function, we further decompose welfare losses into three components based on Equation (37), which expresses welfare losses as a weighted sum of the variances of the employment gap, price inflation, and wage inflation. Figure 3 plots the contributions of the employment gap (red dashed line with diamonds), price inflation (blue dotted line with squares), and wage inflation (green dashed line with triangles).

As the Calvo wage rigidity parameter,  $\theta_w$ , increases, the contribution of price inflation to welfare loss declines monotonically, whereas the contribution of the employment gap rises monotonically. Greater wage rigidity, implemented through quadratic wage adjustment costs, dampens fluctuations in marginal costs, reducing the variance of price inflation and the associated welfare loss. At the same time, lower wage flexibility increases fluctuations in the employment gap, raising the welfare cost associated with employment stabilization. These opposing effects largely offset each other, leaving the wage inflation component as the primary source of the hump-shaped welfare loss function.

The hump-shaped welfare loss associated with wage inflation arises from the interaction between the variance of wage inflation and its weight in the welfare loss function. As  $\theta_w$  increases and wages become more rigid, the variance of wage inflation declines sharply while the weight assigned to wage inflation in the welfare loss function rises exponentially. The interaction of these opposing forces generates a hump-shaped contribution of wage inflation to welfare loss, which in turn plays a central role in shaping the overall welfare loss function. Figure B.2 in the Appendix illustrates this mechanism by plotting the welfare weights together with the variances of the employment gap, price inflation, and wage inflation.

## 5.2 Introducing DNWR

The hump-shaped welfare loss function in Galí and Monacelli (2016) can be overturned when the DNWR constraint binds sufficiently frequently. Figure 4 illustrates the welfare loss function using our baseline model with the DNWR constraint. Each panel plots welfare loss as a function of the Calvo wage rigidity parameter,  $\theta_w$ , across four different shocks: demand shocks (top left), technology shocks (top right), export shocks (bottom left), and world interest rate shocks (bottom right). Once

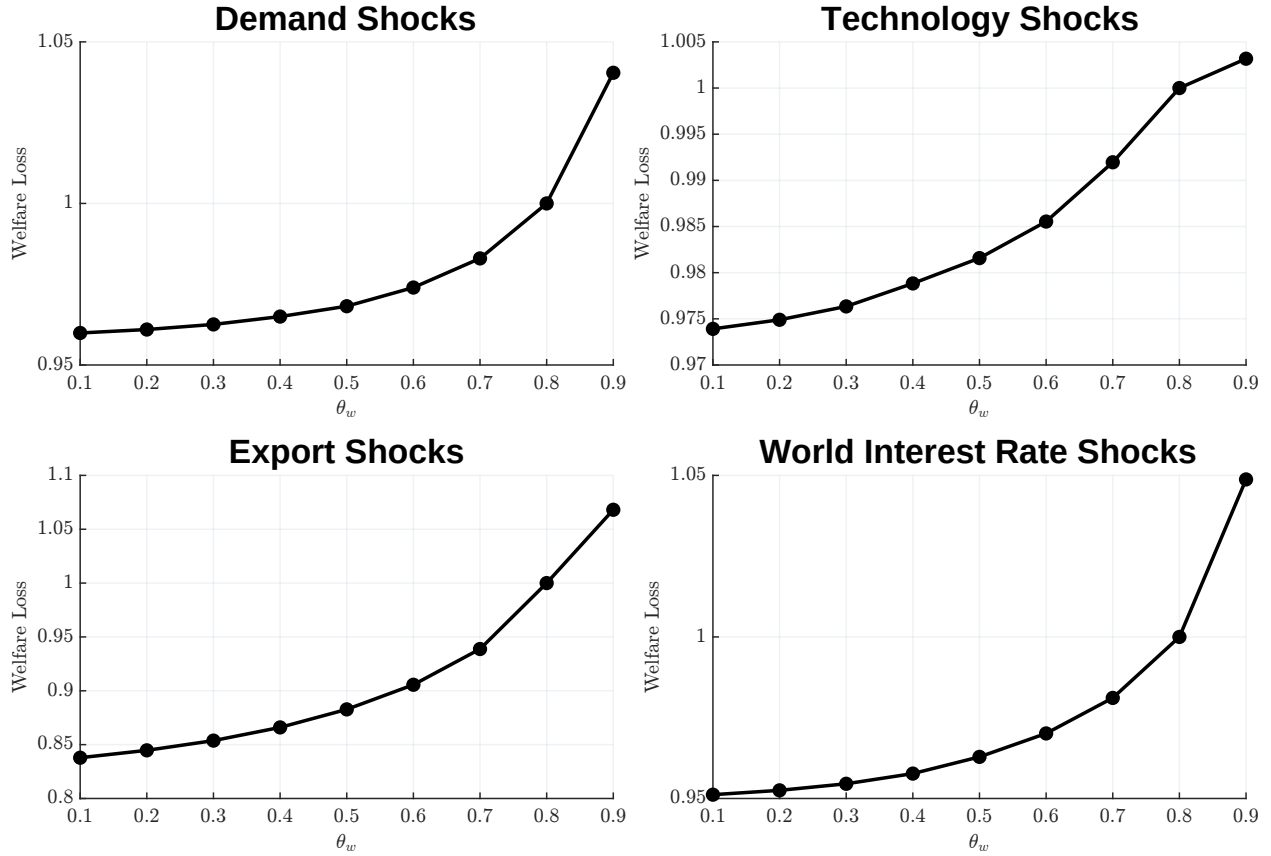


Figure 4: Welfare Loss Across Different Shocks with Varying Calvo Wage Rigidity Parameters with DNWR

Note: This graph plots the welfare loss function based on the model with DNWR, against the Calvo wage rigidity parameter ( $\theta_w$ ) across different types of shocks in a currency union: demand (top-left), technology (top-right), export (bottom left), and world interest rate shocks (bottom right). In each case, the welfare losses are expressed as a ratio relative to those under the baseline setting  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), while all other parameters are fixed at baseline values.

again, welfare losses are expressed as a ratio relative to those under the baseline setting  $\theta_w = 0.8$ .

Compared with the model without DNWR shown in Figure 2, the shape of the welfare loss function changes. Welfare losses now increase monotonically with the Calvo wage rigidity parameter. Consequently, increasing wage flexibility consistently leads to welfare gains. Welfare losses also vary over a narrower range across different degrees of Calvo wage rigidity than in the model without DNWR. These findings suggest that, in environments where DNWR is relevant, additional sources of nominal wage rigidity impose welfare costs by increasing employment fluctuations.

To understand this result, Figure 5 decomposes welfare losses into employment-gap, price-inflation, and wage-inflation components. Relative to the model without DNWR, the contribution of employment-gap fluctuations becomes substantially larger, while the contributions of wage and

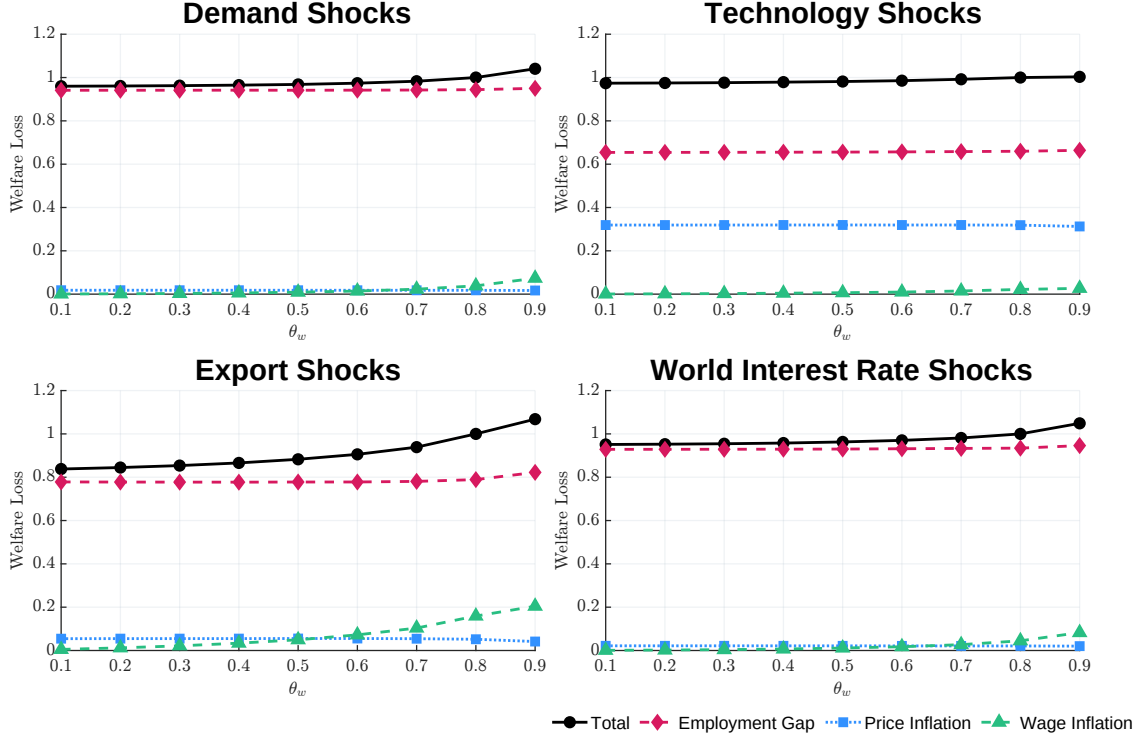


Figure 5: Welfare Loss Decomposition across Different Shocks with Varying Calvo Wage Rigidity Parameters with DNWR

Note: This figure illustrates the decomposition of the welfare loss function based on the model with DNWR against the Calvo wage rigidity parameter ( $\theta_w$ ) across different types of shocks in a currency union: demand (top-left), technology (top-right), export (bottom left), and world interest rate shocks (bottom right). Each line represents the welfare loss attributed to the employment gap (red dashed line with diamonds), price inflation (blue dotted line with squares), and wage inflation (green dashed line with triangles). For each shock, total welfare loss is normalized by the baseline value corresponding to  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), while holding all other parameters fixed. Each component of welfare loss is also normalized by the baseline total welfare loss so that the sum of the components equals total welfare loss for each shock.

price inflation become much smaller. This reflects the fact that DNWR suppresses nominal wage adjustment and thereby dampens both wage and price inflation volatility. Figure B.3 in Appendix B provides further insight by plotting the welfare weights together with the variances of the employment gap, price inflation, and wage inflation.

The intuition is straightforward. In the absence of DNWR, greater wage flexibility shifts adjustment from quantities to prices, reducing employment-gap fluctuations at the cost of higher price and wage inflation volatility. These opposing effects generate the hump-shaped welfare trade-off. When DNWR is present, however, downward wage adjustment is already constrained, substantially compressing both wage and price inflation volatility. As a result, inflation-related distortions become

less important, while employment-gap fluctuations emerge as the dominant source of welfare loss. Consequently, additional wage rigidity primarily affects welfare through its impact on employment fluctuations, causing welfare losses to increase monotonically with Calvo wage rigidity.

### 5.3 State Dependence under DNWR

An important distinction is that DNWR does not simply represent an additional degree of wage rigidity. Calvo wage rigidity affects the frequency of wage adjustment symmetrically, whereas DNWR affects the direction of wage adjustment asymmetrically by restricting wage cuts. As a result, there is no one-to-one mapping between the two frictions that allows the overall degree of wage rigidity to be held constant. In particular, by limiting downward wage adjustment, DNWR substantially compresses wage inflation volatility across all values of  $\theta_w$ , making it difficult to construct a Calvo-only economy with comparable wage dynamics.

To understand the interaction between the two rigidities, Figure B.4 plots welfare losses as a function of both the Calvo wage rigidity parameter  $\theta_w$  and the degree of DNWR,  $\gamma$ . The figure highlights that the welfare implications of symmetric wage rigidity depend on the degree of DNWR.

We first find that stronger DNWR is generally associated with lower welfare losses. The mechanism behind this result is illustrated in Table 3, which reports key moments for economies without and with DNWR under the baseline calibration of  $\theta_w = 0.8$ . We find that introducing DNWR generally reduces welfare losses relative to a model without DNWR. The reason is that DNWR substantially reduces wage and price inflation volatility, thereby lowering the corresponding components of welfare loss. At the same time, DNWR increases employment-gap volatility by shifting part of the adjustment burden toward quantities. Since the welfare weights are unchanged across the two specifications, these differences arise entirely from changes in the volatility of the underlying variables. For the parameter values considered here, the reduction in inflation-related distortions outweighs the increase in employment distortions, resulting in lower overall welfare losses.

We next examine how the strength of DNWR affects the shape of the welfare loss function. As DNWR is weakened (i.e., for  $\gamma < 1$ ), the constraint binds less frequently, wage inflation volatility increases, and the welfare loss function reverts toward the hump-shaped pattern found in Galí and Monacelli (2016). This occurs because, when DNWR becomes less relevant, wage inflation fluctu-

Table 3: Standard Deviations of Employment Gap, Price and Wage Inflation, and Real Wages

|                       | Demand                    |                          | Technology                |                          |
|-----------------------|---------------------------|--------------------------|---------------------------|--------------------------|
|                       | w/o DNWR ( $\gamma = 0$ ) | w/ DNWR ( $\gamma = 1$ ) | w/o DNWR ( $\gamma = 0$ ) | w/ DNWR ( $\gamma = 1$ ) |
| $\sigma(\tilde{n}_t)$ | 1.763%                    | 2.070%                   | 1.837%                    | 1.896%                   |
| $\sigma(\pi_{H,t})$   | 0.069%                    | 0.033%                   | 0.166%                    | 0.155%                   |
| $\sigma(\pi_t^w)$     | 0.102%                    | 0.025%                   | 0.055%                    | 0.021%                   |
| $\sigma(\omega_t)$    | 0.416%                    | 0.104%                   | 0.710%                    | 0.809%                   |
| $\mathbb{L}_t$        | 0.070%                    | 0.051%                   | 0.069%                    | 0.061%                   |

|                       | Export                    |                          | World Interest Rate       |                          |
|-----------------------|---------------------------|--------------------------|---------------------------|--------------------------|
|                       | w/o DNWR ( $\gamma = 0$ ) | w/ DNWR ( $\gamma = 1$ ) | w/o DNWR ( $\gamma = 0$ ) | w/ DNWR ( $\gamma = 1$ ) |
| $\sigma(\tilde{n}_t)$ | 0.513%                    | 0.562%                   | 1.763%                    | 2.080%                   |
| $\sigma(\pi_{H,t})$   | 0.0223%                   | 0.017%                   | 0.069%                    | 0.034%                   |
| $\sigma(\pi_t^w)$     | 0.028%                    | 0.015%                   | 0.102%                    | 0.027%                   |
| $\sigma(\omega_t)$    | 0.088%                    | 0.042%                   | 0.416%                    | 0.089%                   |
| $\mathbb{L}_t$        | 0.006%                    | 0.005%                   | 0.070%                    | 0.052%                   |

Note: This table reports the standard deviation of the employment gap, price inflation, wage inflation, and real wages based on simulated data over 10,000 periods. The results are shown for models without and with DNWR, with Calvo wage rigidity parameter  $\theta_w$  set to 0.8 ( $\phi_w = 864.96$ ) across different types of shocks.

ations increase and the welfare costs associated with wage inflation—which are hump-shaped in  $\theta_w$ —become important relative to employment-gap fluctuations. We further discuss the robustness of our results with respect to  $\gamma$  in Section 5.5.

More generally, the welfare implications of DNWR depend on the frequency with which the constraint binds, which in turn depends on both the size and the sign of shocks. Larger shocks make the DNWR constraint more likely to bind and therefore strengthen the influence of DNWR on welfare outcomes (Figure B.20). As the size of shocks increases, the DNWR constraint binds more frequently and becomes the dominant source of wage rigidity. In this case, variations in the Calvo wage rigidity parameter have a smaller marginal effect because wage adjustment is already largely constrained by DNWR. Consequently, welfare losses become less responsive to changes in  $\theta_w$ , and the welfare loss function becomes flatter.

Similarly, DNWR is primarily relevant following adverse shocks that generate downward pressure on wages (Figure B.21). As a result, welfare losses tend to increase monotonically with Calvo wage rigidity following negative shocks, when the DNWR constraint is frequently binding. By contrast, for positive shocks, the DNWR constraint is largely irrelevant and the hump-shaped welfare trade-off re-emerges. Thus, DNWR does not act as a simple substitute for Calvo wage rigidity. Rather,

it changes the relative importance of inflation and employment fluctuations in determining welfare outcomes, making the welfare implications of wage rigidity state-dependent.

**Precautionary Wage Setting and Solution Method** With both quadratic wage adjustment costs and DNWR present, downward wage adjustments are prohibited, while firms incur adjustment costs when raising wages. Because our framework is solved using a piecewise-linear method, it does not explicitly incorporate precautionary wage-setting behavior—that is, forward-looking wage setters moderating wage increases today in anticipation of future states in which the DNWR constraint may bind. In principle, DNWR alone could generate sluggish upward wage adjustment through this precautionary channel (Elsby, 2009; Daly and Hobijn, 2014).

Our framework captures part of the same mechanism through the presence of explicit costs of upward wage adjustment, which dampen wage increases in response to favorable shocks. In this sense, quadratic wage adjustment costs can be viewed, in part, as a reduced-form representation of sluggish upward wage adjustment associated with the risk of future binding DNWR constraints. Thus, while the model does not explicitly incorporate forward-looking precautionary wage-setting, it captures some of its qualitative implications.

Nonetheless, because the model is not solved globally, certain nonlinearities that a global solution method would capture are excluded from our analysis. In particular, the model does not allow agents to internalize how current wage-setting decisions affect the future probability that the DNWR constraint binds. As a result, higher-order effects associated with precautionary wage-setting behavior may affect the welfare trade-off and the shape of the welfare function. However, the direction of these effects is not obvious. Precautionary wage-setting may dampen upward wage adjustment and thereby increase the importance of employment adjustment. At the same time, it may reduce the likelihood that the DNWR constraint binds in future downturns.

Despite this limitation, the piecewise-linear solution still captures an important asymmetry in employment dynamics—commonly referred to as the “plucking” feature of employment—as emphasized by Dupraz et al. (2021). Under the baseline calibration with  $\theta_w = 0.8$ , the model without DNWR generates an employment skewness of 0.125, while the model with DNWR produces a negative employment skewness of  $-0.04$ . This value is close to the median country-level skewness of the cyclical component of the employment-to-population ratio, which is  $-0.06$  over the period 1991–

2024 for the same sample of countries used in our empirical analysis.<sup>10</sup>

**Heterogeneity by Source of Shocks** The source of shocks also matters for how DNWR affects the composition of welfare loss. In the model with DNWR, the standard deviation of nominal wage inflation becomes similar across different types of shocks, but this is not the case for price inflation. As shown in Table 3, for demand, export, and world interest rate shocks, reduced wage variation leads to more subdued price inflation. By contrast, for technology shocks, productivity fluctuations directly affect marginal costs, generating larger movements in price inflation despite limited nominal wage adjustment. As a result, the dampening effect of DNWR on variance of inflation is substantially weaker for technology shocks than for other shocks.

Therefore, in a currency union, technology shocks also generate much larger fluctuations in real wages than other shocks. Specifically, the standard deviation of real wages reaches 0.8% for technology shocks, compared with 0.04–0.1% for demand, export, and world interest rate shocks. Notably, introducing DNWR reduces real wage volatility for demand, export, and world interest rate shocks, but increases it for technology shocks. This pattern reflects the “inflation greases the wheels” mechanism in the labor market (Tobin, 1972; Card and Hyslop, 1997). Even in the presence of downward nominal wage rigidity, higher inflation volatility generated by technology shocks facilitates real wage adjustment, thereby reducing employment-gap fluctuations. This helps explain why the welfare implications of DNWR vary across shocks. Similar differences in employment responses across business cycle shocks are documented by Jo and Zubairy (2025) in a closed-economy setting.

## 5.4 Monetary Regimes

In Galí and Monacelli (2016), the hump-shaped welfare loss function arises in a currency union. When monetary policy instead responds to domestic inflation – under strict inflation targeting, for example – welfare losses increase monotonically in the degree of Calvo wage rigidity.

We find that this result continues to hold in the presence of DNWR. Figures B.5–B.7 and Figures B.9–B.11 show that welfare losses increase monotonically with Calvo wage rigidity under both strict inflation targeting and a simple Taylor rule ( $i_t = 1.5 \times \pi_{H,t}$ ), regardless of whether DNWR is present.

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<sup>10</sup>We do not calibrate the model to match any specific country-level moment. The country-specific skewness of the cyclical component of the employment-to-population ratio ranges from  $-1.3$  to  $1$ , with a median of  $-0.06$  and a mean of  $-0.2$ .

In other words, DNWR changes the overall shape of the welfare trade-off primarily when monetary policy is constrained.

However, DNWR alters the composition of welfare loss. By limiting downward wage adjustment, DNWR dampens wage and price inflation volatility and shifts a larger share of adjustment toward employment fluctuations. In the model with DNWR, welfare loss is primarily driven by employment-gap fluctuations (see Figures B.8 and B.12 for strict inflation targeting and the Taylor rule, respectively), whereas in the model without DNWR, inflation-related distortions play a relatively larger role (see Figures B.6 and B.10). This highlights that while the monetary regime determines the overall shape of the welfare trade-off, DNWR reshapes the underlying sources of welfare loss.

To further illustrate the role of constrained monetary policy, we consider an environment in which the ZLB constraint binds following a sufficiently large demand shock (see Figure B.13, where  $\sigma_z = 0.05$ ). In the absence of DNWR, the welfare loss function is U-shaped: welfare losses decline as wage rigidity increases, reaching a minimum near  $\theta_w = 0.6$ , before rising as employment fluctuations become more pronounced. Thus, when monetary policy is constrained at the ZLB, greater wage rigidity can improve welfare over part of the parameter range (Billi and Galí, 2020).

When DNWR is introduced, however, welfare losses increase monotonically with Calvo wage rigidity. DNWR suppresses wage and price inflation volatility across all levels of wage rigidity, reducing inflation-related welfare losses. Consequently, employment-gap fluctuations become the dominant source of welfare loss. This mechanism mirrors our findings in the currency-union regime, where DNWR reduces inflation-related distortions and shifts the burden of adjustment toward employment fluctuations.

The ZLB exercise treats monetary policy as completely constrained. However, in practice, central banks at the ZLB also use unconventional monetary policy (Wu and Xia, 2016, 2020). Wu and Zhang (2019a,b) show that such policies can be represented in a New Keynesian model by an effective interest rate that tracks a shadow rate rather than the policy rate at its effective lower bound. In our implementation, the shadow rate is the notional rate consistent with strict inflation targeting, and we assume full pass-through of the shadow rate to the effective rate.

Figure B.14 reports welfare losses as a function of the Calvo wage rigidity parameter under effective unconventional monetary policy (UMP), using the same demand-shock process ( $\sigma_z = 0.05$ )

as in Figure B.13. Without DNWR, UMP restores the monetary response to inflation and removes the U-shaped pattern: welfare losses increase monotonically with wage rigidity, as in the economy without a ZLB constraint. Greater wage flexibility therefore improves welfare. With DNWR, the welfare loss function remains monotonically increasing under UMP, as it is under the strict ZLB.<sup>11</sup>

Taken together, these exercises show that the welfare effects of wage rigidity depend on the scope for monetary stabilization. When monetary policy cannot respond effectively to local conditions, as in a currency union or at a binding ZLB without effective unconventional policy, greater wage rigidity can improve welfare in the absence of DNWR. When DNWR binds sufficiently frequently, however, it suppresses wage and price inflation volatility and shifts adjustment toward employment, causing welfare losses to increase with Calvo wage rigidity.

## 5.5 Robustness

Our main result is that welfare losses increase with Calvo wage rigidity under the baseline calibration. In this section, we examine how this result depends on the strength of DNWR, the size and composition of shocks, the steady-state wage inflation, and other model parameters.

We first consider alternative values of the DNWR parameter,  $\gamma$ . In the model,  $\gamma$  governs the lower bound on nominal wage growth when the DNWR constraint binds. Our calibration exercise shows that, for most countries, the implied value of  $\gamma$  is close to one on average (Table A.3). However, the implied value of  $\gamma$  for Greece is close to  $\gamma = 0.99$  during the Great Recession, consistent with its unusually high incidence of wage cuts. To assess robustness, we therefore consider a weaker form of DNWR by setting  $\gamma = 0.99$ .

Figure B.15 plots the resulting welfare loss functions. Because a lower value of  $\gamma$  implies that the DNWR constraint binds less frequently, we also consider a larger shock size ( $\sigma_z = 0.3$ ) chosen so that the simulated downturn is comparable in magnitude to the Greek episode during the Great Recession.<sup>12</sup> We find that the upward-sloping relationship between welfare loss and Calvo wage rigidity is preserved over most of the parameter space when the DNWR constraint binds sufficiently

<sup>11</sup>We also note that even when the currency union as a whole is at the ZLB and adopts UMP, union-wide accommodation moves the union-wide interest rate rather than providing a country-specific policy instrument. As long as the exchange rate remains pegged, union-wide UMP does not restore country-specific stabilization, and the welfare loss function in the currency union would remain hump-shaped without DNWR, as in Galí and Monacelli (2016).

<sup>12</sup>Under this calibration employment falls by 27%. Greece's unemployment rate reached 27.7% in 2013. We present this large-shock case as a sensitivity exercise.

frequently.

The key mechanism operates through the frequency with which the DNWR constraint binds. Figure B.16 reports this frequency and shows that the shape of the welfare loss function depends critically on the relevance of the DNWR constraint. When  $\gamma$  is lower and shocks are relatively small, the DNWR constraint binds less frequently. This effect is particularly pronounced at high levels of Calvo wage rigidity, where the average size of wage adjustment is small and the DNWR constraint is therefore less likely to bind. Consistent with this mechanism, Figure B.16 shows that the frequency of binding declines as  $\theta_w$  increases, making DNWR progressively less relevant at high levels of wage rigidity.

When DNWR binds less frequently, the economy behaves similarly to the standard model without DNWR, and the hump-shaped welfare trade-off of Galí and Monacelli (2016) re-emerges. By contrast, when shocks are sufficiently large, the DNWR constraint binds frequently and employment-gap fluctuations become the dominant source of welfare loss (Figure B.17). In this case, wage and price inflation volatility are strongly compressed, so additional Calvo wage rigidity primarily affects welfare through its impact on employment fluctuations rather than inflation stabilization. As a result, welfare losses increase monotonically with Calvo wage rigidity over most of the parameter space. These results reinforce our central finding that the welfare implications of DNWR are state-dependent and governed by the frequency with which the constraint binds.

The same mechanism explains the results under alternative steady-state wage inflation rates. Figure B.18 shows that higher steady-state wage inflation causes the hump-shaped welfare loss function to re-emerge. Consistent with this result, Figure B.19 shows that higher steady-state wage inflation weakens the relevance of DNWR by reducing the likelihood that nominal wage cuts are required, thereby lowering the frequency with which the DNWR constraint binds. As DNWR becomes less relevant, the economy behaves more like the standard model without DNWR, and the hump-shaped welfare trade-off reappears. When we increase the size of shocks, however, the DNWR constraint binds more frequently, and the monotonic relationship between welfare loss and Calvo wage rigidity is restored over most of the parameter space.

We also examine robustness with respect to labor supply elasticity, the elasticity of substitution across labor types, and the degree of openness. Figures B.22, B.23, and B.24 show that these parameters affect the quantitative magnitude of welfare losses but do not alter the main qualita-

tive mechanism. Whenever the DNWR constraint is sufficiently relevant, employment-gap volatility remains the dominant source of welfare loss and welfare losses increase with Calvo wage rigidity.

Overall, the robustness exercises show that the welfare implications of DNWR are inherently state-dependent. The key determinant is not simply the presence of DNWR itself, but the frequency with which the constraint binds. This frequency is governed by the interaction between the strength of DNWR, the size and composition of shocks, the steady-state wage inflation, and the degree of symmetric wage rigidity. Across all specifications, the monotonic welfare result emerges whenever the DNWR constraint binds sufficiently frequently. Based on the analysis we report, declining segments of the welfare loss function arise only in regions where the constraint binds infrequently – on the order of 20% of quarters or less – while the increasing relationship prevails wherever the constraint binds more frequently (Figures B.15, B.18, B.16, B.19). We interpret this as a description of our simulations rather than as a sharp threshold.

## 6 Aggregate Demand and Supply under DNWR

To illustrate how DNWR affects business cycle fluctuations in a currency union, we recover aggregate demand (AD) and aggregate supply (AS) relationships from simulated data.<sup>13</sup> Demand-side shocks—including demand, export, and world interest rate shocks—shift the AD curve while leaving the AS relationship unchanged, allowing us to trace out the AS curve. Conversely, supply shocks shift the AS curve while leaving the AD relationship unchanged, allowing us to recover the AD curve.

The slopes of the AD and AS curves play an important role in determining the transmission of shocks. Shocks shift either the AD or AS curve, and the resulting real effects depend on the slopes of these curves. Therefore, understanding how DNWR affects the AD and AS curves helps clarify how shock propagation changes in the presence of DNWR. The left panel of Figure 6 plots the AD–AS relationships without DNWR, while the right panel presents the model with DNWR.

We find that DNWR substantially flattens the AS curve, or Phillips curve. This flattening of the Phillips curve holds broadly across alternative parameter specifications. A higher degree of DNWR or larger shocks leads to a more pronounced flattening of the AS curve (see Figure C.26). When employment declines and the DNWR constraint binds, marginal costs decline less, thereby

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<sup>13</sup>We simulate the model for 10,000 periods for each shock, setting  $\theta_w = 0.8$  (corresponding to  $\phi_w = 864.96$ )

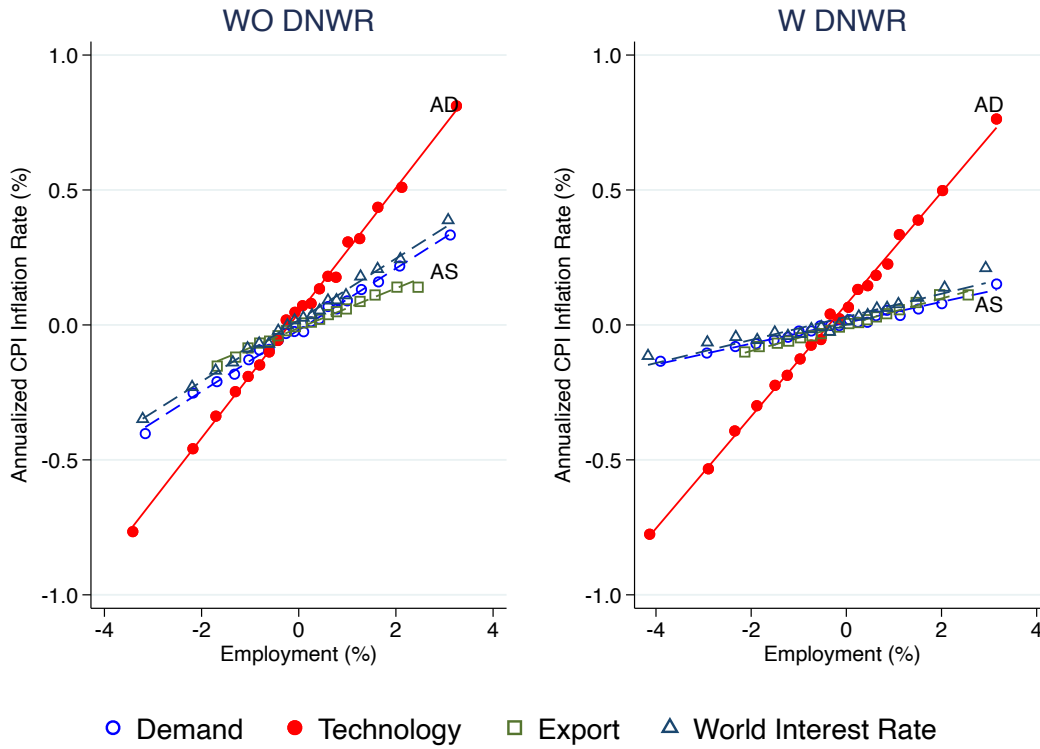


Figure 6: AD-AS Curves without and with DNWR

Note: This graph plots the simulated annualized CPI inflation rate against employment, comparing the model without DNWR (left panel) and with DNWR (right panel) in response to four different shocks: demand (hollow circles), supply (solid circles), export (hollow squares), and world interest rate (hollow triangles) shocks. We set  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ) and simulated 10,000 observations for each shock. In the left panel (without DNWR), the slope coefficients are 0.0285, 0.0582, 0.0189, and 0.0285 for demand, technology, export, and world interest rate shocks, respectively. In the right panel (with DNWR), the slope coefficients are 0.00982, 0.0522, 0.0127, and 0.00996 for demand, technology, export, and world interest rate shocks, respectively.

flattening the slope of the AS curve. This mechanism has also been documented in the closed-economy literature (Daly and Hobijn, 2014; Mineyama, 2023). We show that it also operates in a small open economy within a currency union.

As a result, policy interventions or shocks that shift the AD curve have larger effects on employment and output when DNWR is present. Consistent with this interpretation, Figure C.25 in the Appendix shows that negative demand shocks generate larger employment declines in the model with DNWR within a currency union. By contrast, under a strict inflation-targeting regime, monetary policy responds endogenously to subdued inflationary pressures associated with DNWR, largely offsetting changes in shock propagation following demand shocks.

In a currency union where monetary policy cannot respond to country-specific conditions, the AD curve is upward-sloping regardless of the presence of DNWR. The intuition is that a favorable supply shock lowers marginal costs and reduces inflationary pressures. Under an independent monetary policy regime, the central bank would typically lower nominal interest rates to support aggregate demand. In a currency union, however, monetary policy does not respond to domestic economic conditions. As a result, lower inflation raises the real interest rate, effectively tightening monetary conditions. The increase in the real interest rate suppresses aggregate demand, generating the paradoxical outcome that a positive supply shock can lead to a short-run contraction in employment. This mechanism has also been emphasized in NK models during zero lower bound episodes when monetary policy is unresponsive (Eggertsson, 2010; Eggertsson and Krugman, 2012; Eggertsson et al., 2014). The presence of DNWR also modestly affects the AD relationship through its effect on inflation dynamics, although the quantitative impact is limited.<sup>14</sup>

## 7 Conclusion

This paper revisits the welfare implications of wage sluggishness in a currency union by incorporating DNWR into a small open economy NK framework. Using evidence from the WDN surveys, we document that wage freezes are widespread across European countries and become particularly prevalent during recessions. Consistent with the presence of DNWR, countries experiencing larger employment declines exhibit a higher incidence of wage freezes relative to wage cuts.

Motivated by these empirical patterns, we extend the framework of Galí and Monacelli (2016) to allow for DNWR and examine its welfare implications. In contrast to previous studies that emphasize the stabilizing role of wage sluggishness through lower wage and price inflation volatility, we find that welfare losses increase monotonically with Calvo wage rigidity once DNWR binds sufficiently frequently. DNWR substantially compresses wage and price inflation volatility and shifts adjustment toward employment. As a result, the welfare costs associated with employment fluctuations dominate the benefits of reduced inflation volatility, leading welfare losses to increase monotonically with

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<sup>14</sup>When the DNWR constraint binds, wage inflation—and therefore price inflation—respond less strongly to changes in economic conditions. As a result, inflation expectations and real interest rates move less for a given change in employment. Since nominal interest rates are fixed within a currency union, this mechanism modestly flattens the AD curve (see Figure C.26). Although this flattening of the AD curve can amplify the real effects of supply shocks, the quantitative impact remains limited. Consistent with this result, the right panel of Figure C.25 shows that the employment responses to a technology shock with and without DNWR are quantitatively very similar.

Calvo wage rigidity.

More generally, the welfare implications of DNWR are state-dependent and governed by the frequency with which the constraint binds. Our analysis also highlights that these effects depend on the monetary policy regime. The monotonic welfare result emerges in environments where stabilization policy is constrained, such as currency unions or periods in which the zero lower bound binds. We further show that DNWR flattens the AS (Phillips) curve while having only a limited effect on the AD curve, implying larger employment responses to demand shocks when the constraint binds. Taken together, these findings underscore the importance of accounting for downward wage rigidities when evaluating labor market institutions and macroeconomic stabilization policies.

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# Online Appendix

## A Additional Empirical Evidence

Table A.1 presents the percentage of workers who experienced wage increases, freezes, and cuts across 13 European Union (EU) countries between 2007 and 2013. Table A.2 reports regression results based on Equation (1), using the unemployment rate as the key explanatory variable. The results show that increases in unemployment are associated with a substantially larger rise in the share of workers experiencing wage freezes than in the share experiencing wage cuts. Figure A.1 plots the change in the difference between the share of workers experiencing wage freezes and those experiencing wage cuts from 2007 to 2013, against changes in unemployment rate over the same period.

Table A.3 reports the implied values of  $\gamma$  for the countries in our sample using aggregate wage data. We use two measures from the OECD: annual wages and labor compensation per hour.<sup>15</sup>

In the model,  $\gamma = W_t/W_{t-1}$  when the DNWR constraint binds (i.e., when  $\Delta_t > 0$ ). Periods with  $\Delta_t > 0$  correspond to states in which the marginal benefit of relaxing the DNWR constraint is positive, which we associate with periods of rising unemployment.

Based on this implication, to calibrate  $\gamma$  at a quarterly frequency, we first compute the average annual wage growth rate during periods when aggregate unemployment is increasing. Since the wage data are available at an annual frequency, we convert them to quarterly frequency using

$$\gamma_Q^4 = \gamma_A, \tag{A.1}$$

where  $\gamma_Q$  denotes the wage growth rate when the DNWR constraint binds at the quarterly frequency, and  $\gamma_A$  denotes the corresponding wage growth rate at the annual frequency.

Using the longest available sample, we find that the implied  $\gamma$  equals or exceeds one for most countries, indicating that nominal wages, on average, continue to grow even during downturns when

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<sup>15</sup>Average annual wages are defined as the annual earnings per full-time equivalent employee in the total economy. Data are available at <https://www.oecd.org/en/data/indicators/average-annual-wages.html>. Labor compensation per hour worked reflects the average cost of labor per hour paid by employers. Data are available at <https://www.oecd.org/en/data/indicators/labour-compensation-per-hour-worked.html>.

Table A.1: Wage Freezes and Wage Cuts across European Countries

| Country | Year | Percent of Workers |             |          | Country     | Year | Percent of Workers |             |          |
|---------|------|--------------------|-------------|----------|-------------|------|--------------------|-------------|----------|
|         |      | Wage Increase      | Wage Freeze | Wage Cut |             |      | Wage Increase      | Wage Freeze | Wage Cut |
| Austria | 2007 | 91.36              | 8.27        | 0.37     | Italy       | 2007 | 98.66              | 1.19        | 0.15     |
| Austria | 2009 | 98.71              | 1.27        | 0.02     | Italy       | 2009 | 68.95              | 31.00       | 0.04     |
| Austria | 2010 | 96.47              | 2.47        | 1.05     | Italy       | 2010 | 93.26              | 6.46        | 0.29     |
| Austria | 2011 | 97.92              | 2.07        | 0.02     | Italy       | 2011 | 92.83              | 6.80        | 0.38     |
| Austria | 2012 | 96.54              | 3.34        | 0.12     | Italy       | 2012 | 90.32              | 9.07        | 0.61     |
| Austria | 2013 | 97.74              | 2.04        | 0.22     | Italy       | 2013 | 88.40              | 10.20       | 1.41     |
| Czechia | 2007 | 85.57              | 12.87       | 1.55     | Lithuania   | 2007 | 95.21              | 3.87        | 0.92     |
| Czechia | 2009 | 50.84              | 49.13       | 0.03     | Lithuania   | 2010 | 69.90              | 25.85       | 4.25     |
| Czechia | 2010 | 81.26              | 16.76       | 1.98     | Lithuania   | 2011 | 80.56              | 18.60       | 0.84     |
| Czechia | 2011 | 84.01              | 14.32       | 1.67     | Lithuania   | 2012 | 85.76              | 13.05       | 1.19     |
| Czechia | 2012 | 84.07              | 14.28       | 1.65     | Lithuania   | 2013 | 87.84              | 11.29       | 0.88     |
| Czechia | 2013 | 86.83              | 11.28       | 1.89     |             |      |                    |             |          |
| Estonia | 2007 | 91.71              | 8.08        | 0.21     | Netherlands | 2007 | 85.99              | 13.83       | 0.19     |
| Estonia | 2009 | 42.67              | 56.94       | 0.39     | Netherlands | 2009 | 87.38              | 12.58       | 0.04     |
| Estonia | 2010 | 86.37              | 8.35        | 5.28     | Netherlands | 2010 | 92.77              | 7.16        | 0.07     |
| Estonia | 2011 | 92.16              | 6.18        | 1.66     | Netherlands | 2011 | 81.27              | 18.66       | 0.06     |
| Estonia | 2012 | 96.24              | 2.97        | 0.79     | Netherlands | 2012 | 79.78              | 20.08       | 0.15     |
| Estonia | 2013 | 97.60              | 2.15        | 0.25     | Netherlands | 2013 | 75.93              | 23.51       | 0.56     |
| France  | 2007 | 93.91              | 4.99        | 1.10     | Portugal    | 2007 | 92.12              | 7.72        | 0.16     |
| France  | 2009 | 17.47              | 82.48       | 0.05     | Portugal    | 2010 | 77.41              | 22.20       | 0.39     |
| France  | 2010 | 93.00              | 7.00        | 0.00     | Portugal    | 2011 | 67.99              | 29.39       | 2.61     |
| France  | 2011 | 92.94              | 7.02        | 0.04     | Portugal    | 2012 | 61.78              | 35.42       | 2.80     |
| France  | 2012 | 90.73              | 9.24        | 0.03     | Portugal    | 2013 | 63.36              | 34.20       | 2.44     |
| France  | 2013 | 85.76              | 14.12       | 0.11     |             |      |                    |             |          |
| Greece  | 2007 |                    | 9.82        |          | Slovenia    | 2007 | 96.41              | 2.40        | 1.19     |
| Greece  | 2010 | 58.53              | 34.84       | 6.63     | Slovenia    | 2010 | 95.37              | 4.63        | 0.00     |
| Greece  | 2011 | 54.70              | 31.19       | 14.12    | Slovenia    | 2011 | 94.45              | 5.55        | 0.00     |
| Greece  | 2012 | 46.26              | 24.70       | 29.04    | Slovenia    | 2012 | 93.28              | 6.72        | 0.00     |
| Greece  | 2013 | 51.08              | 28.04       | 20.88    | Slovenia    | 2013 | 93.76              | 6.24        | 0.00     |
| Hungary | 2007 | 96.56              | 3.17        | 0.27     | Spain       | 2007 | 98.83              | 1.16        | 0.01     |
| Hungary | 2010 | 97.61              | 2.20        | 0.19     | Spain       | 2009 | 78.05              | 21.94       | 0.01     |
| Hungary | 2011 | 95.95              | 3.75        | 0.30     | Spain       | 2010 | 84.16              | 14.71       | 1.13     |
| Hungary | 2012 | 95.41              | 4.18        | 0.41     | Spain       | 2011 | 77.97              | 21.34       | 0.69     |
| Hungary | 2013 | 97.64              | 2.28        | 0.09     | Spain       | 2012 | 67.28              | 30.38       | 2.34     |
|         |      |                    |             |          | Spain       | 2013 | 67.86              | 24.92       | 7.22     |
| Ireland | 2007 | 92.74              | 6.88        | 0.37     |             |      |                    |             |          |
| Ireland | 2010 | 40.00              | 47.36       | 12.64    |             |      |                    |             |          |
| Ireland | 2011 | 44.01              | 49.13       | 6.86     |             |      |                    |             |          |
| Ireland | 2012 | 47.80              | 46.29       | 5.92     |             |      |                    |             |          |
| Ireland | 2013 | 49.24              | 42.81       | 7.95     |             |      |                    |             |          |

Table A.2: Spike at zero and the share of cuts with unemployment rate

|                      | (1)<br>Share of Freezes<br>$\Delta W = 0$ | (2)<br>Share of Cuts<br>$\Delta W < 0$ | (3)<br>Share of Freezes Minus<br>Share of Cuts |
|----------------------|-------------------------------------------|----------------------------------------|------------------------------------------------|
| Unemployment Rate    | 1.651***<br>(0.412)                       | 0.533***<br>(0.130)                    | 1.136*<br>(0.564)                              |
| Inflation Rate       | 1.522<br>(1.313)                          | -0.600<br>(0.504)                      | 2.147<br>(1.676)                               |
| Country-Fixed Effect | Yes                                       | Yes                                    | Yes                                            |
| Time-Fixed Effect    | Yes                                       | Yes                                    | Yes                                            |
| Observations         | 72                                        | 71                                     | 71                                             |
| Adjusted $R^2$       | 0.57                                      | 0.83                                   | 0.47                                           |

Data Source: WEO, WDI, and WDN. Sample Period: 2007, 2009, 2010-2013. Sample Countries: Austria, Belgium, Bulgaria, Cyprus, Czech Republic, Germany, Estonia, Spain, France, Greece, Croatia, Hungary, Ireland, Italy, Lithuania, Luxembourg, Latvia, Malta, Netherlands, Poland, Portugal, Romania, Slovenia, and United Kingdom. Country-level clustered robust standard errors are shown in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

unemployment is increasing.

Focusing on the subsample corresponding to the WDN data, Greece has the lowest implied values, at slightly above 0.99. This is broadly consistent with the WDN evidence, which shows that Greece experienced the highest incidence of wage cuts—29% in 2012 and 21% in 2013—while other countries report wage cuts affecting less than 5% of workers. During this period, Greece also experienced a sharp increase in unemployment, reaching 24.7% in 2012 and 27.7% in 2013. The second-highest incidence of wage cuts is observed in Ireland, where 12.6% of workers experienced wage cuts in 2010, when unemployment reached 15%. However, even in this case, the implied  $\gamma$  remains close to one.

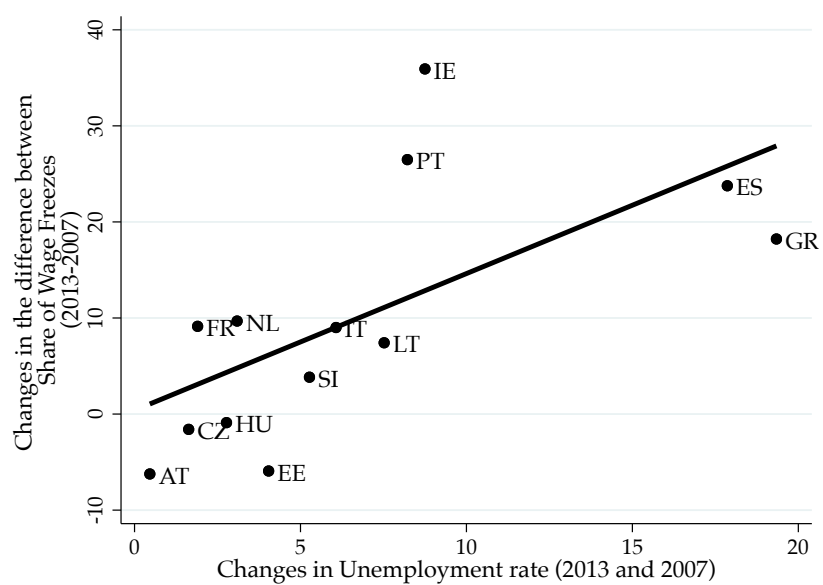
As a robustness check, we repeat the analysis using  $\gamma = 0.99$ , which allows for up to a 4% annual wage decline and corresponds to the lower bound of  $\gamma$  observed in the data. One caveat is that replicating the Greek experience requires shocks large enough to generate unemployment rates exceeding 25%. To account for this, we also increase the size of the baseline shock accordingly. Our main results remain robust under this alternative calibration.

Table A.3: Implied Values of the DNWR Parameter ( $\gamma$ ), Adjusted for Long-Run Productivity Growth

| Sample Period<br>Country | Annual wage |           | Labor compensation per hour |           |
|--------------------------|-------------|-----------|-----------------------------|-----------|
|                          | 1991–2024   | 2007–2013 | 1996–2022                   | 2007–2013 |
| Austria                  | 1.006       | 1.004     | 1.005                       | 1.007     |
| Czechia                  | 1.010       | 1.002     | 1.010                       | 1.000     |
| Estonia                  | 1.018       | 1.003     | 1.023                       | 1.008     |
| France                   | 1.005       | 1.004     | 1.005                       | 1.005     |
| Greece                   | 1.004       | 0.991     | 1.005                       | 0.993     |
| Hungary                  | 1.016       | 1.005     | 1.012                       | 1.002     |
| Ireland                  | 1.000       | 0.997     | 1.002                       | 0.998     |
| Italy                    | 1.006       | 1.003     | 1.004                       | 1.005     |
| Lithuania                | 1.006       | 0.998     | 1.003                       | 0.998     |
| Netherlands              | 1.006       | 1.003     | 1.005                       | 1.004     |
| Portugal                 | 1.006       | 1.001     | 1.006                       | 1.001     |
| Slovenia                 | 1.007       | 1.000     | 1.006                       | 0.999     |
| Spain                    | 1.008       | 1.005     | 1.006                       | 1.005     |
| (Unweighted) Median      | 1.006       | 1.003     | 1.005                       | 1.002     |

*Notes:* Each entry reports the implied quarterly value of  $\gamma$ , computed as  $\exp(m/4)$ , where  $m$  is the average over rising-unemployment years ( $\Delta u_t > 0$ ) of annual log nominal wage growth net of the country's long-run total factor productivity growth. Data sources are from the OECD. Average annual wages are available at <https://www.oecd.org/en/data/indicators/average-annual-wages.html>. Labor compensation per hour worked is available at <https://www.oecd.org/en/data/indicators/labour-compensation-per-hour-worked.html>. Annual wage data are available for most countries from 1991–2024, except for Czechia, Estonia, Greece, Hungary, Lithuania, Portugal, and Slovenia, for which data are available from 1996 onward. Labor compensation per hour worked is available for 1996–2022. TFP growth is the country average over 1991 - 2024 of total-economy TFP from AMECO (variable ZVGDF, European Commission DG ECFIN), available from 1996 for Czechia, Estonia, Hungary, Lithuania, and Slovenia. Columns (1) and (3) report the implied quarterly values of  $\gamma$  using the longest available sample. Columns (2) and (4) report the implied quarterly values of  $\gamma$  for the subsample corresponding to the WDN data.

Figure A.1: Change in Spike at Zero and Change in Unemployment Rate



Data Source: WEO and WDN. Notes: The figure plots the change in the share of workers experiencing wage freezes against the change in the unemployment rate between 2007 and 2013 across countries in the EU, with a fitted line. Each dot on the graph represents a country, with the respective abbreviations: AT for Austria, CZ for the Czech Republic, EE for Estonia, ES for Spain, FR for France, GR for Greece, HU for Hungary, IE for Ireland, IT for Italy, LT for Lithuania, NL for the Netherlands, PT for Portugal, and SI for Slovenia.

## B Additional Welfare Results

### B.1 Decomposition of Welfare Loss

Figure B.2 and Figure B.3 plot the coefficients on the employment gap, price inflation, and wage inflation in the welfare loss function given by Equation (37), together with the corresponding variances of these variables in models without and with DNWR across four different shocks. Importantly, the coefficients are unchanged across specifications. Introducing DNWR reduces the volatility of wage and price inflation while increasing the volatility of the employment gap.

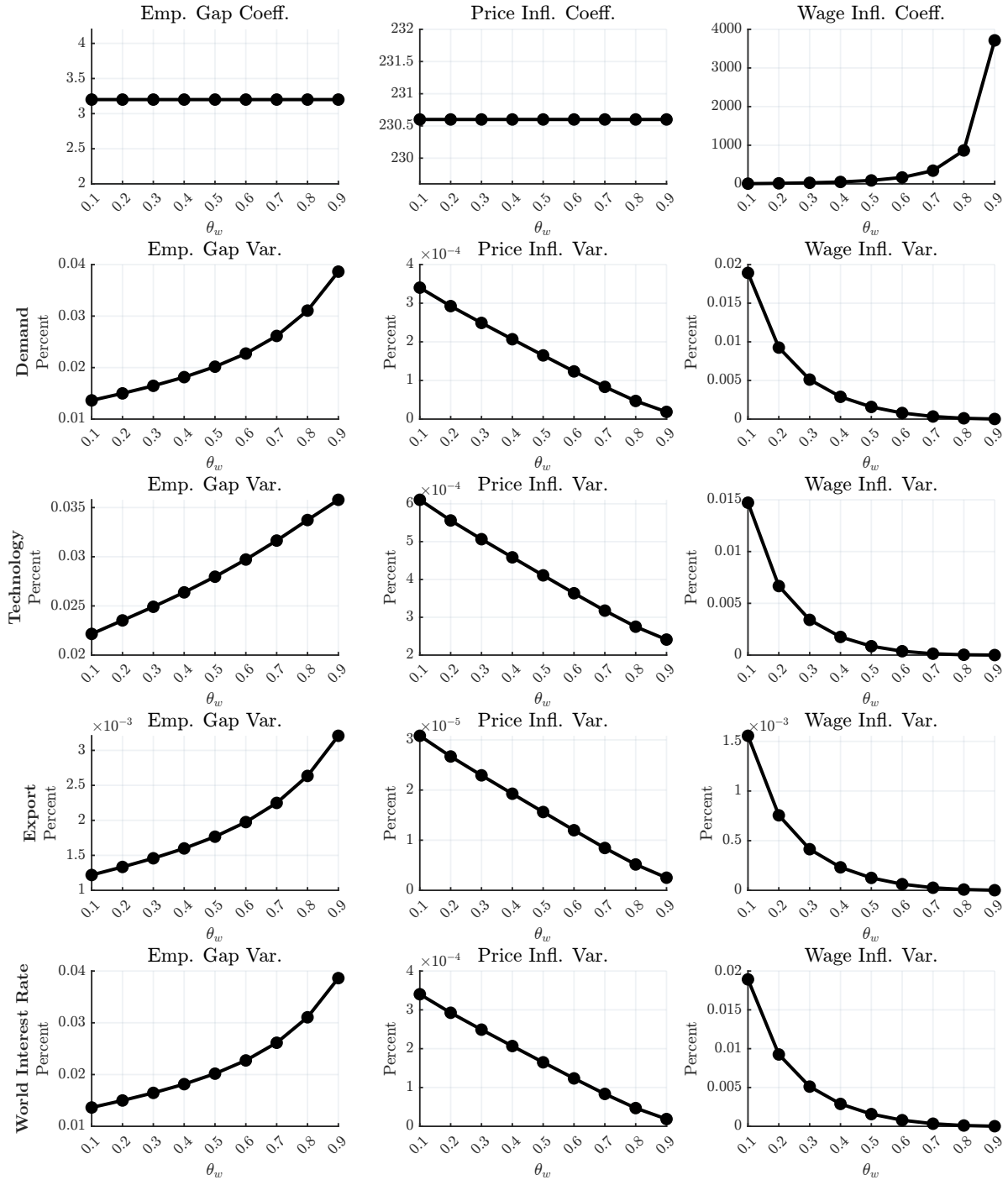


Figure B.2: Components of Welfare Loss across Shocks without DNWR as a Function of Calvo Wage Rigidity

Note: This figure illustrates the components of welfare loss in a model without DNWR, calculated based on Equation (37), across four different shocks as a function of the Calvo wage rigidity parameter,  $\theta_w$ . The first row displays the weights on the employment gap, price inflation, and wage inflation, while the second through fifth rows depict the variances of the employment gap, price inflation, and wage inflation for each of the four different shocks.

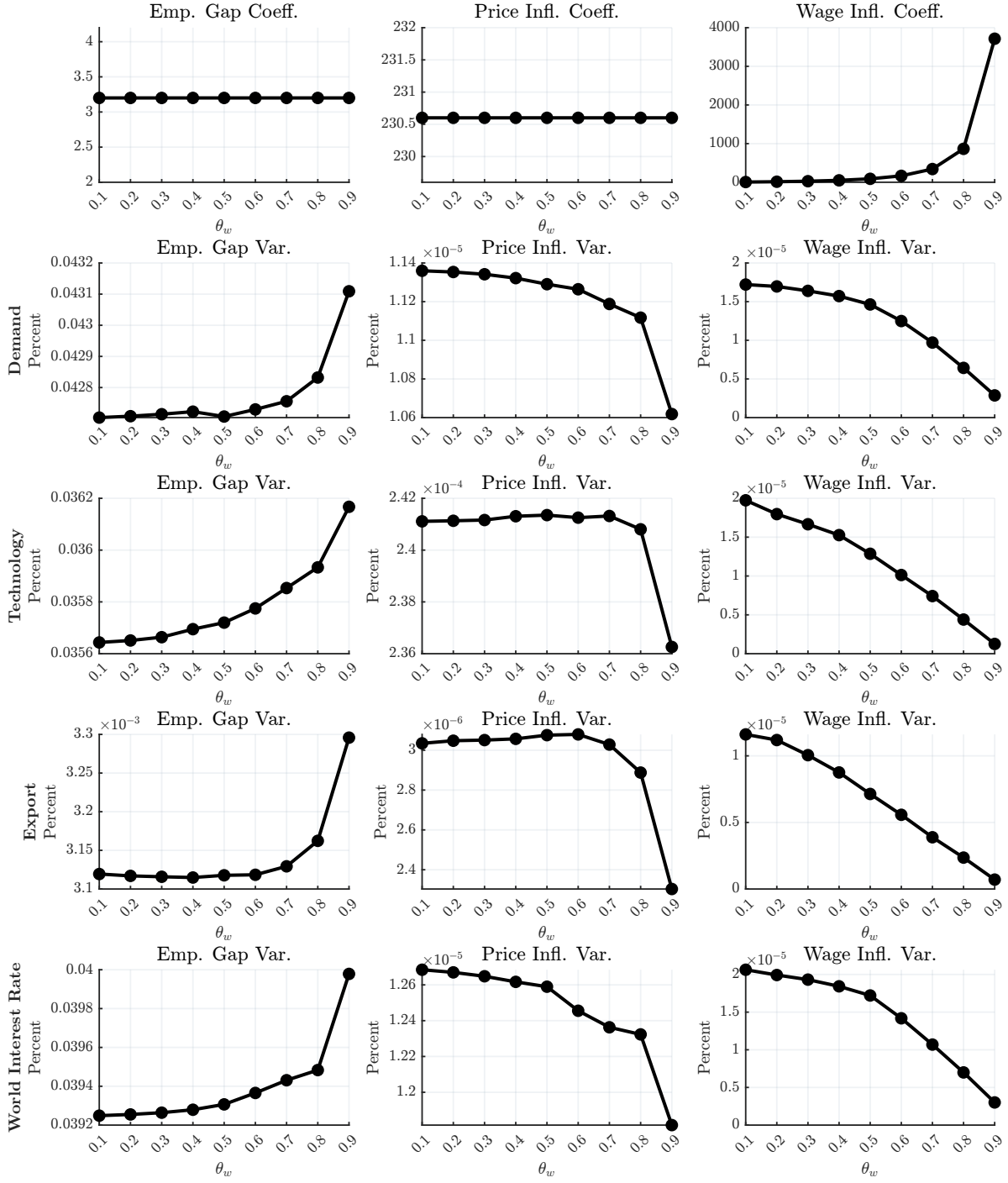


Figure B.3: Components of Welfare Loss across Shocks with DNWR as a Function of Calvo Wage Rigidity

Note: This figure illustrates the components of welfare loss in a model with DNWR, calculated based on Equation (37), across four different shocks as a function of the Calvo wage rigidity parameter,  $\theta_w$ . The first row displays the weights on the employment gap, price inflation, and wage inflation, while the second through fifth rows depict the variances of the employment gap, price inflation, and wage inflation for each of the four different shocks.

## B.2 Interaction between Calvo wage rigidity and DNWR

Figure B.4 illustrates how welfare losses vary jointly with the Calvo wage rigidity parameter ( $\theta_w$ ) and the DNWR parameter ( $\gamma$ ). The figure highlights how the welfare implications of Calvo wage rigidity depend on the strength of DNWR.

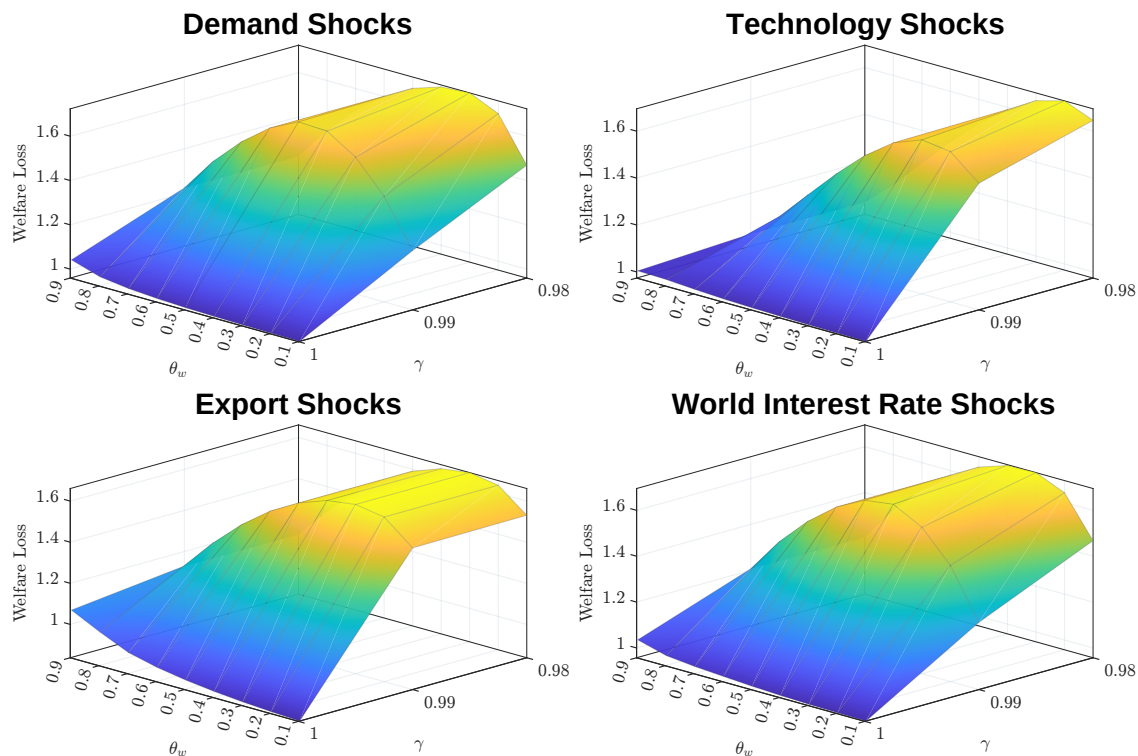


Figure B.4: Welfare Loss across Different Shocks as a Function of Calvo Wage Rigidity and DNWR

Note: The figure plots welfare losses as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) and the DNWR parameter ( $\gamma$ ) for demand (top-left), technology (top-right), export (bottom-left), and world interest rate shocks (bottom-right). Welfare losses are normalized to one at the baseline calibration,  $\theta_w = 0.8$  and  $\gamma = 1$ . The shock size is set to  $\sigma = 0.01$ .

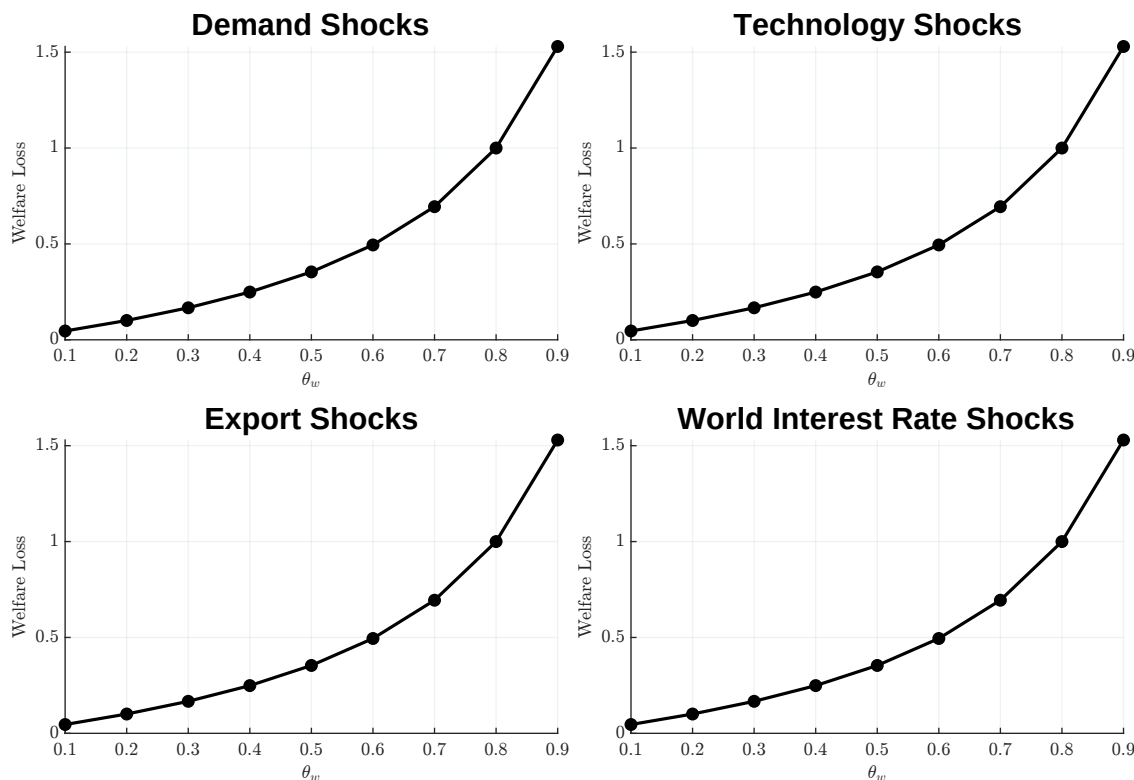


Figure B.5: Welfare Loss under Strict Inflation Targeting *without* DNWR

*Note:* This figure plots the welfare loss function in the model without DNWR as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) across different shocks under strict inflation targeting: demand (top-left), technology (top-right), export (bottom-left), and world interest rate (bottom-right). In each case, welfare losses are normalized by their value at the baseline setting  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), with all other parameters held fixed.

### B.3 Alternative Monetary Policy Regimes

Figures B.5–B.8 report welfare losses and their decomposition under strict inflation targeting, with- and without DNWR. Figures B.9–B.12 present the corresponding results under a Taylor rule.

Consistent with the main text, welfare losses increase monotonically with Calvo wage rigidity under both monetary policy regimes. However, the composition of welfare loss differs across specifications. In the presence of DNWR, employment-gap fluctuations account for a larger share of welfare losses, whereas inflation-related distortions play a relatively larger role in the model without DNWR.

Figure B.13 reports welfare losses under a binding zero lower bound (ZLB). To generate ZLB episodes, we simulate a large demand shock with  $\sigma_z = 0.05$ .

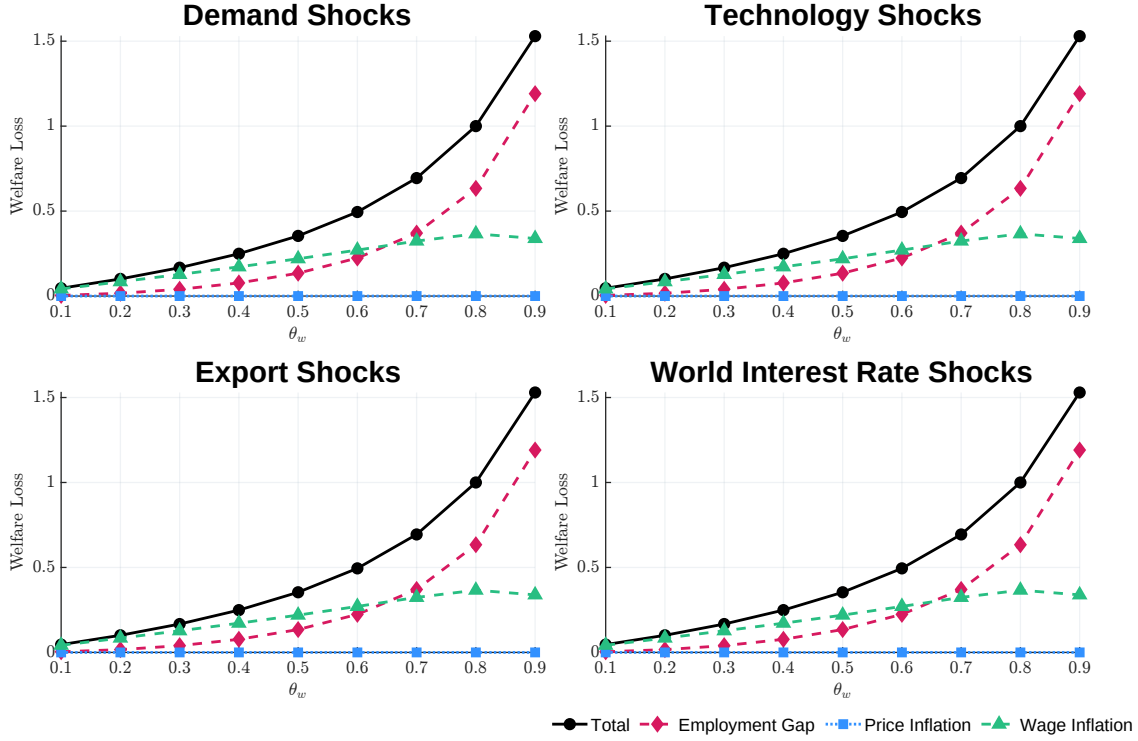


Figure B.6: Decomposition of Welfare Loss under Strict Inflation Targeting *without* DNWR

Note: This figure illustrates the decomposition of the welfare loss function based on the model without DNWR against the Calvo wage rigidity parameter ( $\theta_w$ ) across different types of shocks under strict inflation targeting: demand (top-left), technology (top-right), export (bottom left), and world interest rate shocks (bottom right). Each line represents the welfare loss attributed to the employment gap (red dashed line with diamonds), price inflation (blue dotted line with squares), and wage inflation (green dashed line with triangles). For each shock, total welfare loss is normalized by the baseline value corresponding to  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), while holding all other parameters fixed. Each component is normalized by baseline total welfare loss so that the sum of the components equals total welfare loss in each panel.

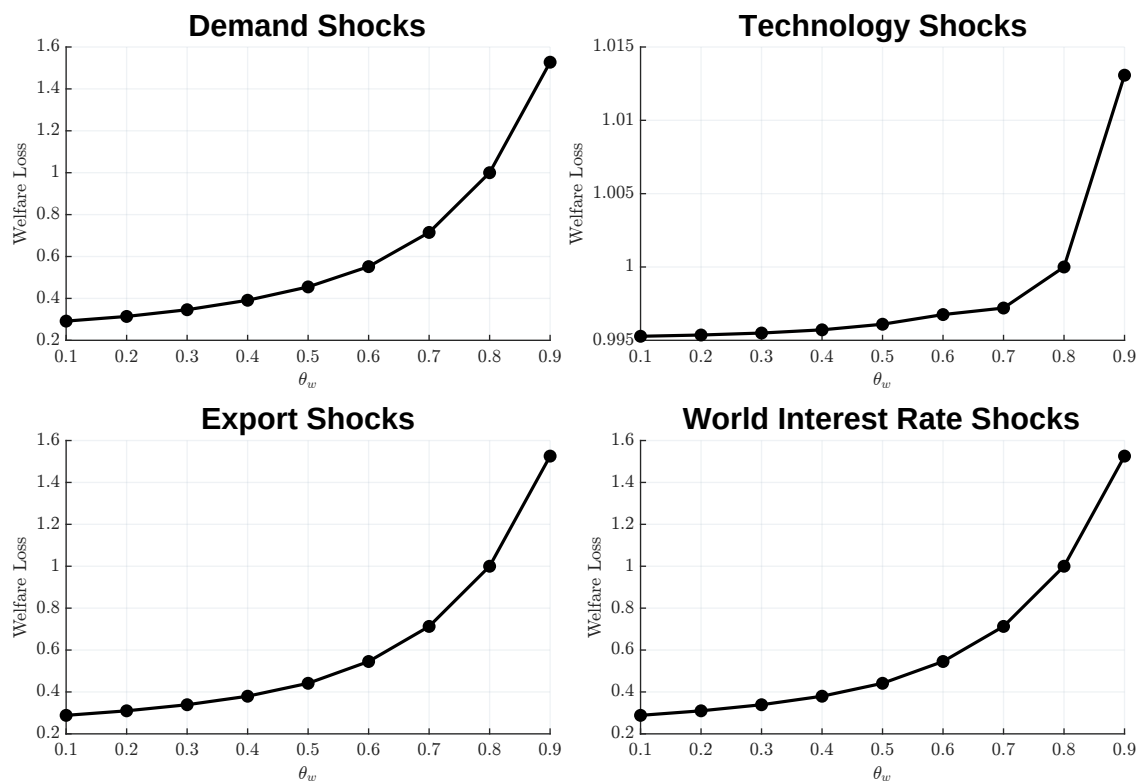


Figure B.7: Welfare Loss under Strict Inflation Targeting *with* DNWR

*Note:* This figure plots the welfare loss function in the model with DNWR as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) across different shocks under strict inflation targeting: demand (top-left), technology (top-right), export (bottom-left), and world interest rate (bottom-right). In each case, welfare losses are normalized by their value at the baseline setting  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), with all other parameters held fixed.

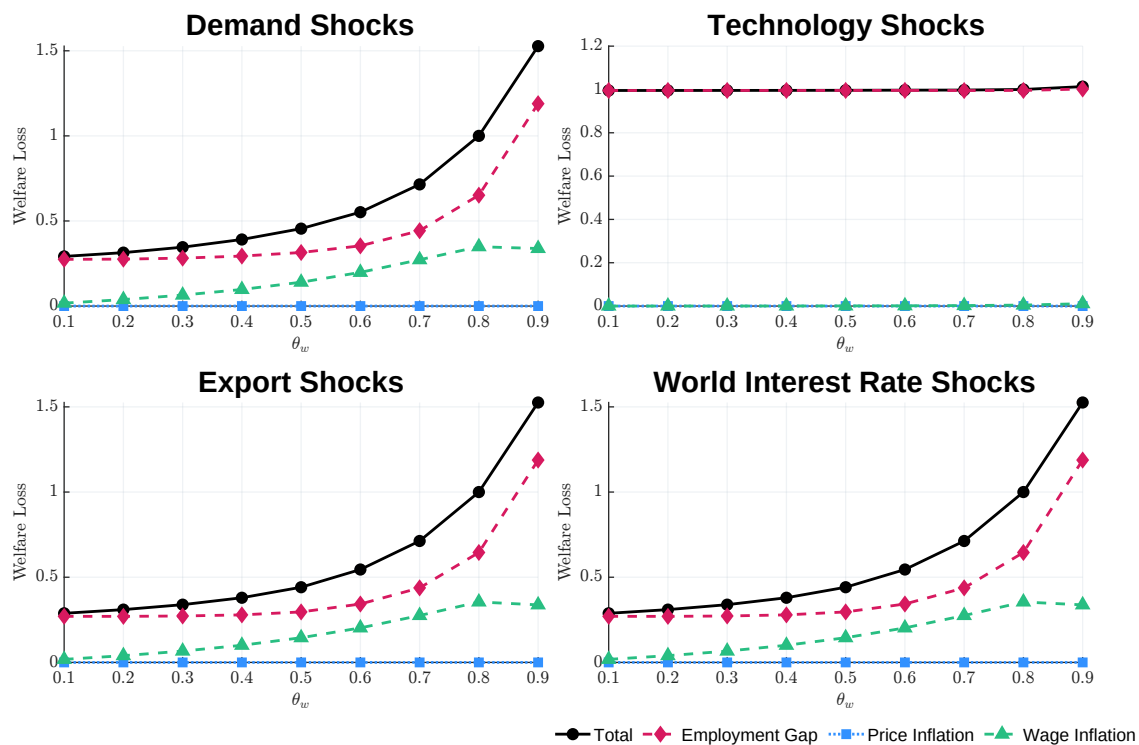


Figure B.8: Welfare Loss Decomposition under Strict Inflation Targeting *with* DNWR

Note: This figure illustrates the decomposition of the welfare loss function based on the model with DNWR against the Calvo wage rigidity parameter ( $\theta_w$ ) across different types of shocks under strict inflation targeting: demand (top-left), technology (top-right), export (bottom left), and world interest rate shocks (bottom right). Each line represents the welfare loss attributed to the employment gap (red dashed line with diamonds), price inflation (blue dotted line with squares), and wage inflation (green dashed line with triangles). For each shock, total welfare loss is normalized by the baseline value corresponding to  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), while holding all other parameters fixed. Each component is normalized by baseline total welfare loss so that the sum of the components equals total welfare loss in each panel.

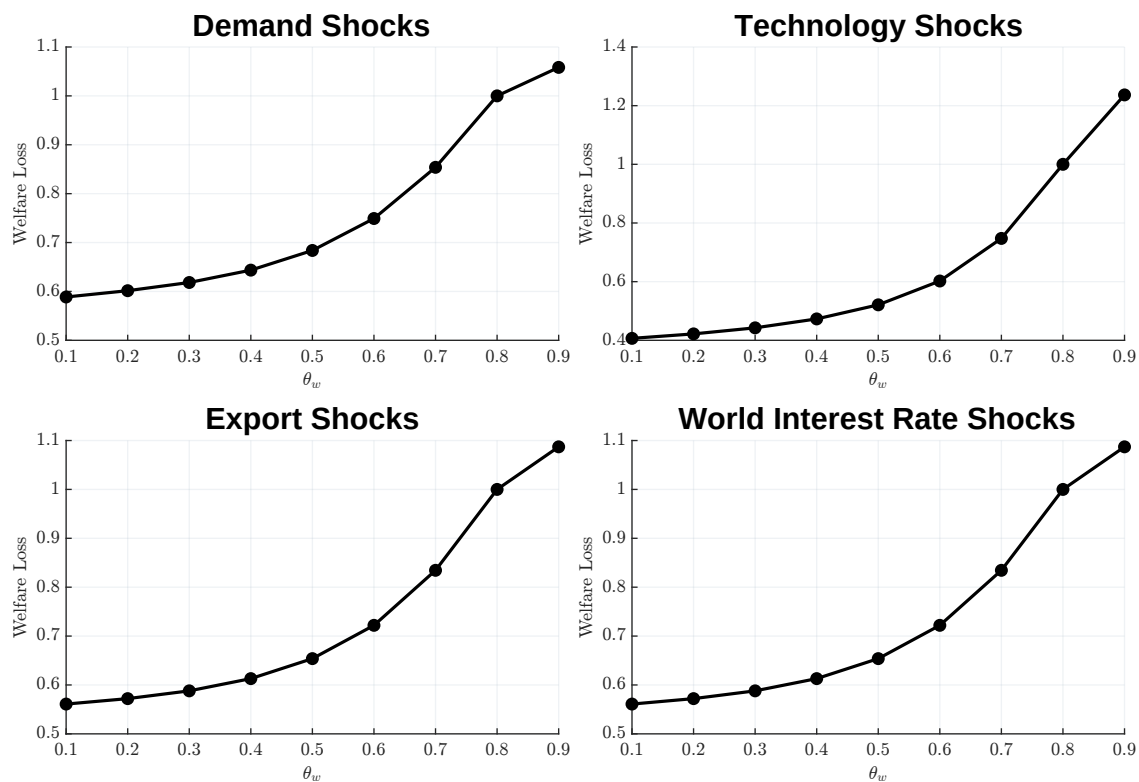


Figure B.9: Welfare Loss under a Taylor Rule *without* DNWR

*Note:* This figure plots the welfare loss function in the model without DNWR as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) across different shocks under a Taylor rule: demand (top-left), technology (top-right), export (bottom-left), and world interest rate (bottom-right). In each case, welfare losses are normalized by their value at the baseline setting  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), with all other parameters held fixed.

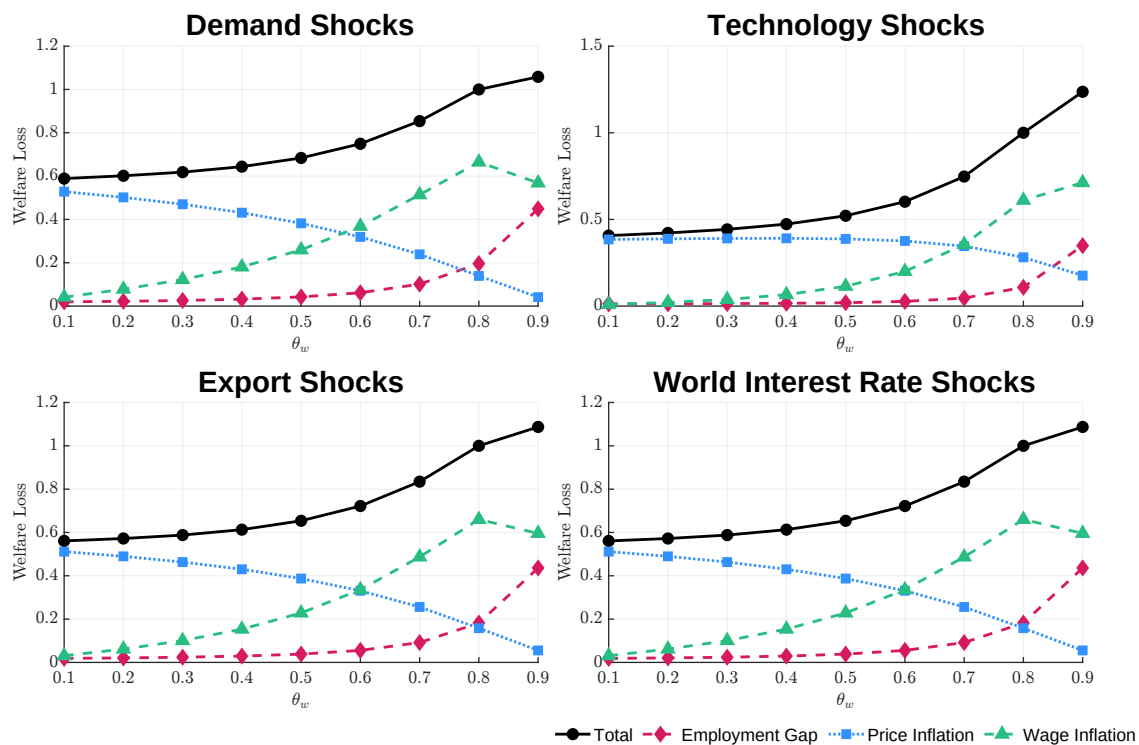


Figure B.10: Welfare Loss Decomposition under a Taylor Rule *without* DNWR

Note: This figure illustrates the decomposition of the welfare loss function based on the model without DNWR against the Calvo wage rigidity parameter ( $\theta_w$ ) across different types of shocks under a Taylor rule: demand (top-left), technology (top-right), export (bottom left), and world interest rate shocks (bottom right). Each line represents the welfare loss attributed to the employment gap (red dashed line with diamonds), price inflation (blue dotted line with squares), and wage inflation (green dashed line with triangles). For each shock, total welfare loss is normalized by the baseline value corresponding to  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), while holding all other parameters fixed. Each component is normalized by baseline total welfare loss so that the sum of the components equals total welfare loss in each panel.

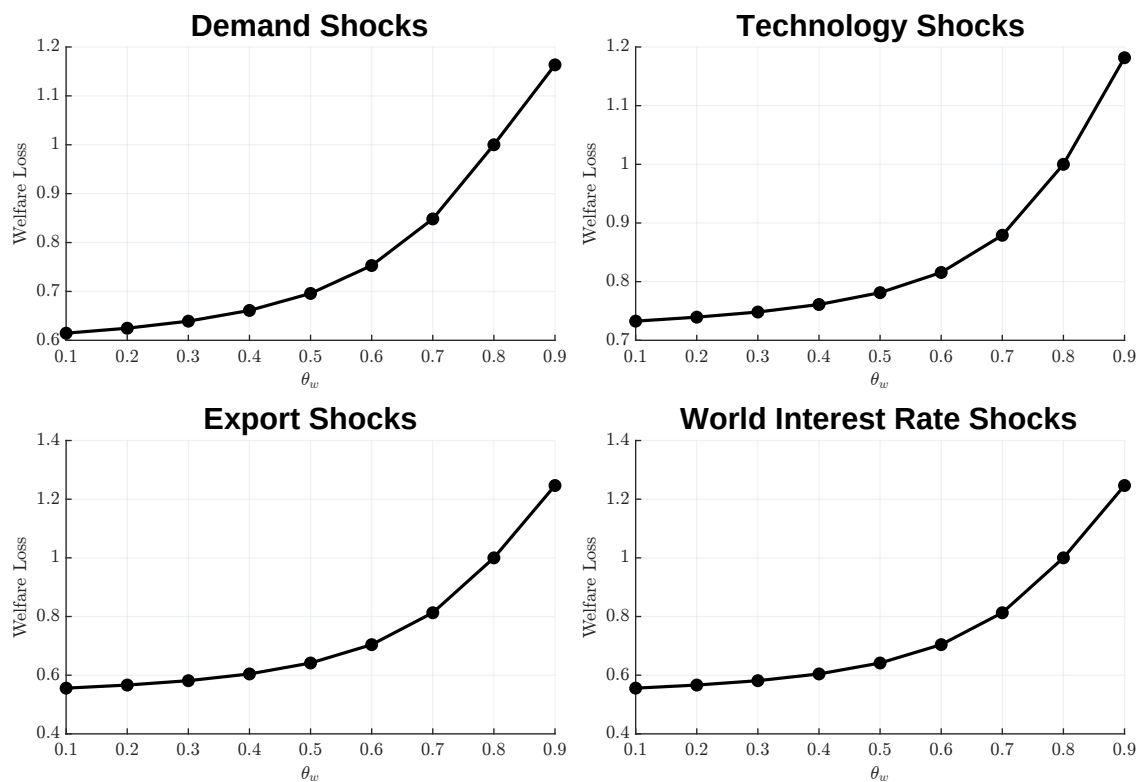


Figure B.11: Welfare Loss under a Taylor Rule *with* DNWR

*Note:* This figure plots the welfare loss function in the model with DNWR as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) across different shocks under a Taylor rule: demand (top-left), technology (top-right), export (bottom-left), and world interest rate (bottom-right). In each case, welfare losses are normalized by their value at the baseline setting  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), with all other parameters held fixed.

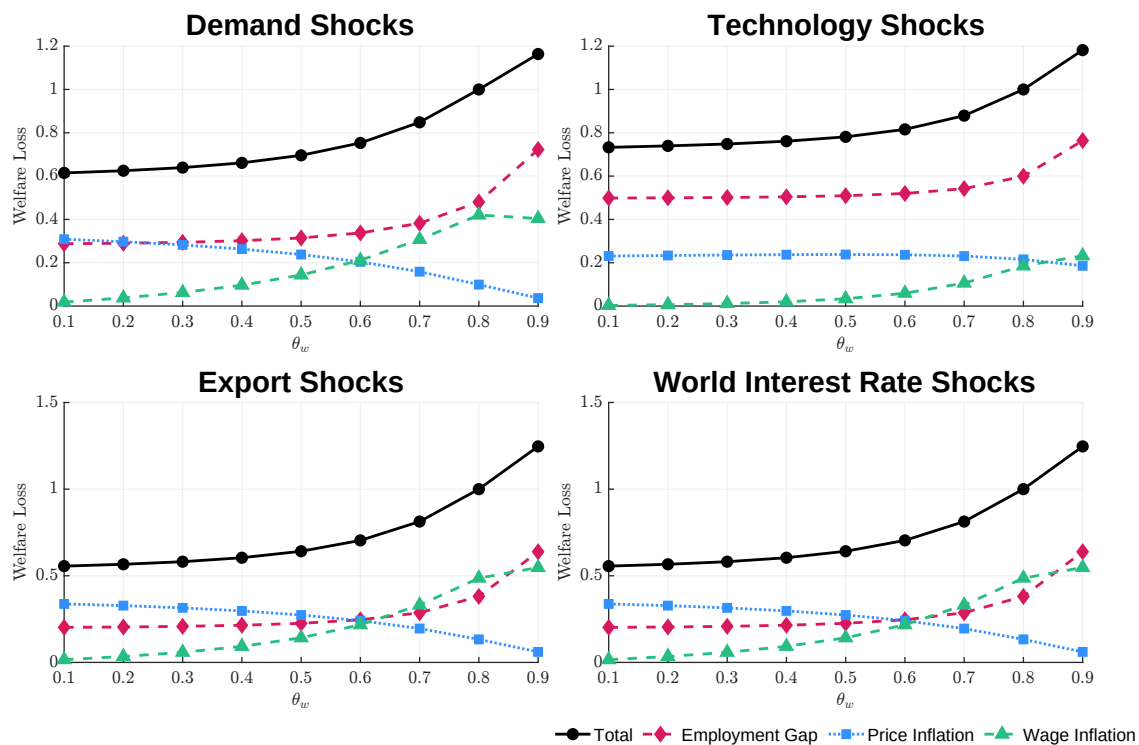


Figure B.12: Welfare Loss Decomposition under a Taylor Rule *with* DNWR

Note: This figure illustrates the decomposition of the welfare loss function based on the model with DNWR against the Calvo wage rigidity parameter ( $\theta_w$ ) across different types of shocks under a Taylor rule: demand (top-left), technology (top-right), export (bottom left), and world interest rate shocks (bottom right). Each line represents the welfare loss attributed to the employment gap (red dashed line with diamonds), price inflation (blue dotted line with squares), and wage inflation (green dashed line with triangles). For each shock, total welfare loss is normalized by the baseline value corresponding to  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), while holding all other parameters fixed. Each component is normalized by baseline total welfare loss so that the sum of the components equals total welfare loss in each panel.

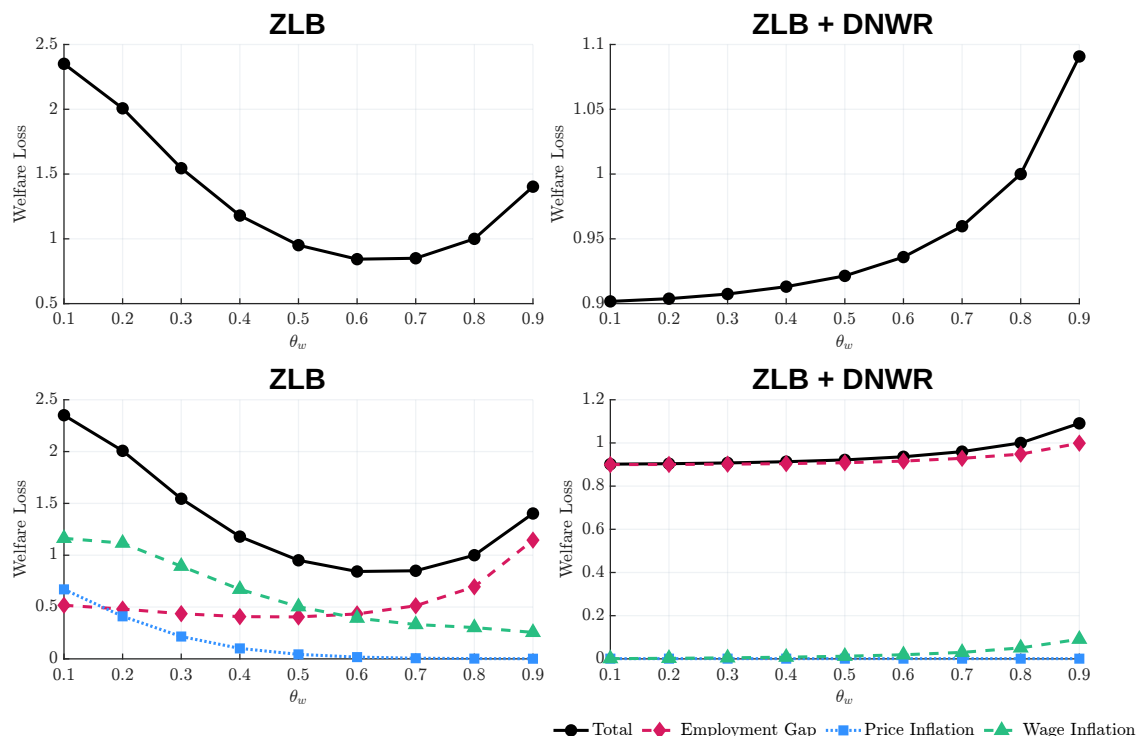


Figure B.13: Welfare Loss under Strict Inflation Targeting with a Binding ZLB

Note: The figure plots welfare losses and their decomposition in response to a large demand shock ( $\sigma_z = 0.05$ ) under strict inflation targeting with a binding zero lower bound (ZLB). The left column reports results for the model without DNWR, while the right column reports results for the model with DNWR. The top panels plot total welfare losses as a function of the Calvo wage rigidity parameter ( $\theta_w$ ). The bottom panels decompose welfare losses into contributions from the employment gap, price inflation, and wage inflation. The black solid line with circles represents total welfare loss; the remaining lines represent the welfare loss attributed to the employment gap (red dashed line with diamonds), price inflation (blue dotted line with squares), and wage inflation (green dashed line with triangles). Welfare losses in each panel are normalized by that panel's value at the baseline calibration,  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), so the figures compare the shapes of the loss functions. The decomposition components are normalized by baseline total welfare loss so that their sum equals total welfare loss.

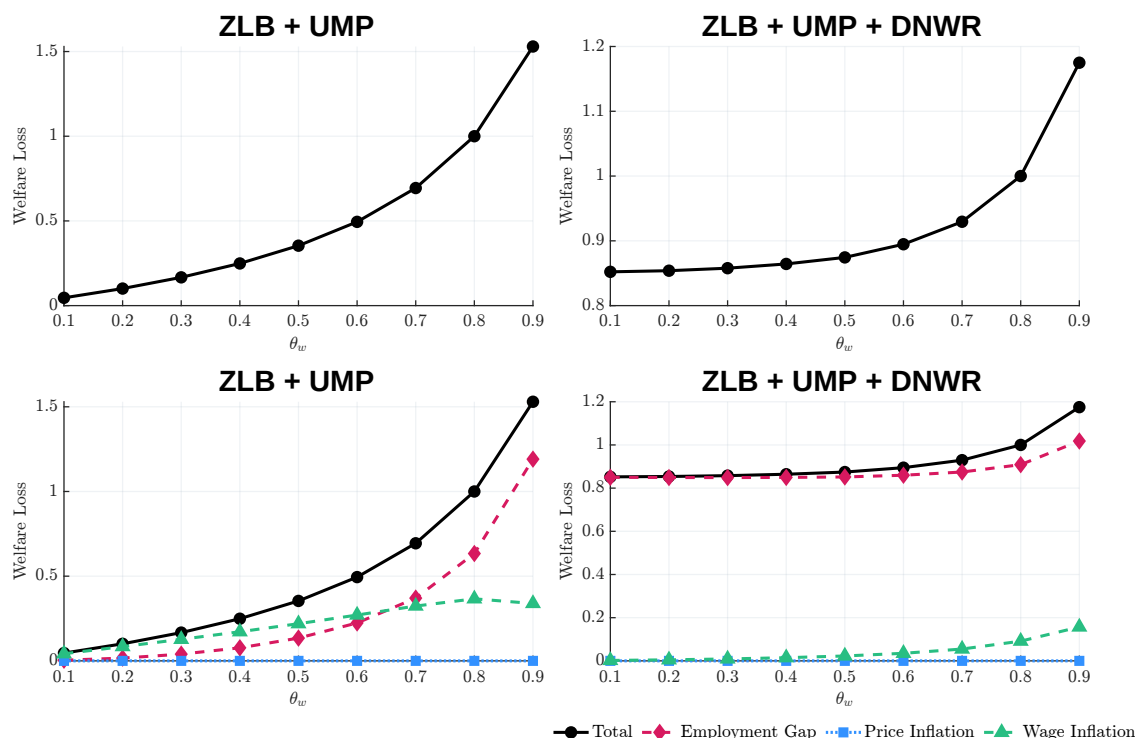


Figure B.14: Welfare Loss under Strict Inflation Targeting with Unconventional Monetary Policy

Note: The figure plots welfare losses and their decomposition in response to a large demand shock ( $\sigma_z = 0.05$ ) under strict inflation targeting with unconventional monetary policy. Unconventional monetary policy is modeled as full pass-through of the shadow rate—the notional rate consistent with strict inflation targeting—to the effective interest rate. The left column reports results for the model without DNWR, while the right column reports results for the model with DNWR. The top panels plot total welfare losses as a function of the Calvo wage rigidity parameter ( $\theta_w$ ). The bottom panels decompose welfare losses into contributions from the employment gap, price inflation, and wage inflation. The black solid line with circles represents total welfare loss; the remaining lines represent the welfare loss attributed to the employment gap (red dashed line with diamonds), price inflation (blue dotted line with squares), and wage inflation (green dashed line with triangles). Welfare losses in each panel are normalized by that panel's value at the baseline calibration,  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), so the figures compare the shapes of the loss functions.

## B.4 Alternative Parameter Specification

Figure B.15 shows the welfare loss function with respect to Calvo wage rigidity parameter ( $\theta_w$ ) with less strict DNWR constraint ( $\gamma = 0.99$ ) by varying the size of shock. For each shock, welfare losses are expressed as ratios relative to the baseline value at  $\theta_w = 0.8$ . Figure B.16 plots corresponding frequency of binding DNWR for each parameter specification. Figure B.17 plots decomposition of welfare loss for each shock with  $\gamma = 0.99$

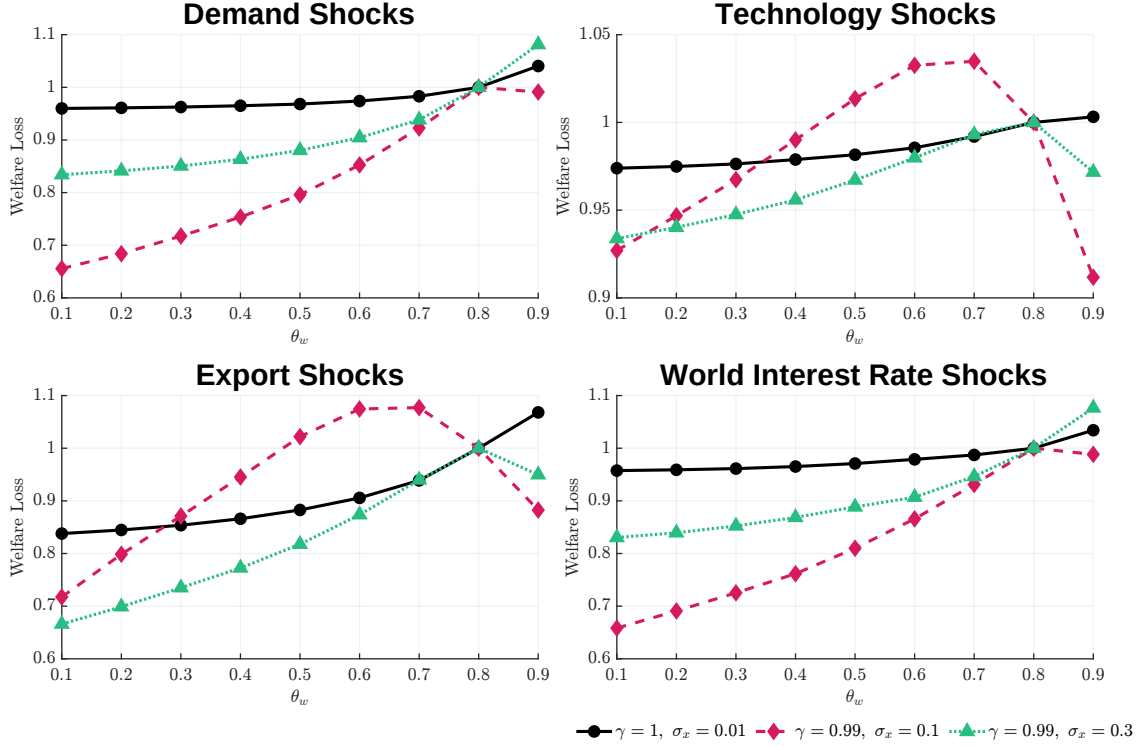


Figure B.15: Welfare Loss under a Weaker DNWR Constraint ( $\gamma = 0.99$ )

Note: This figure plots welfare losses in the model with DNWR as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) for four types of shocks: demand (top-left), technology (top-right), export (bottom-left), and world interest rate shocks (bottom-right). The black solid line with circles corresponds to the baseline case with  $\gamma = 1$  and  $\sigma_z = 0.01$ . The red dashed line with diamonds corresponds to  $\gamma = 0.99$  and  $\sigma_z = 0.1$ , while the green dotted line with triangles corresponds to  $\gamma = 0.99$  and  $\sigma_z = 0.3$ . In all cases, welfare losses are normalized by their value at the baseline calibration,  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), with all other parameters held at their baseline values.

Figure B.18 plots welfare losses as a function of the Calvo wage rigidity parameter under alternative values of steady-state wage inflation,  $\Pi^w$ . Figure B.19 reports the corresponding frequency with which the DNWR constraint binds.

Higher steady-state wage inflation reduces the likelihood that nominal wage cuts are required

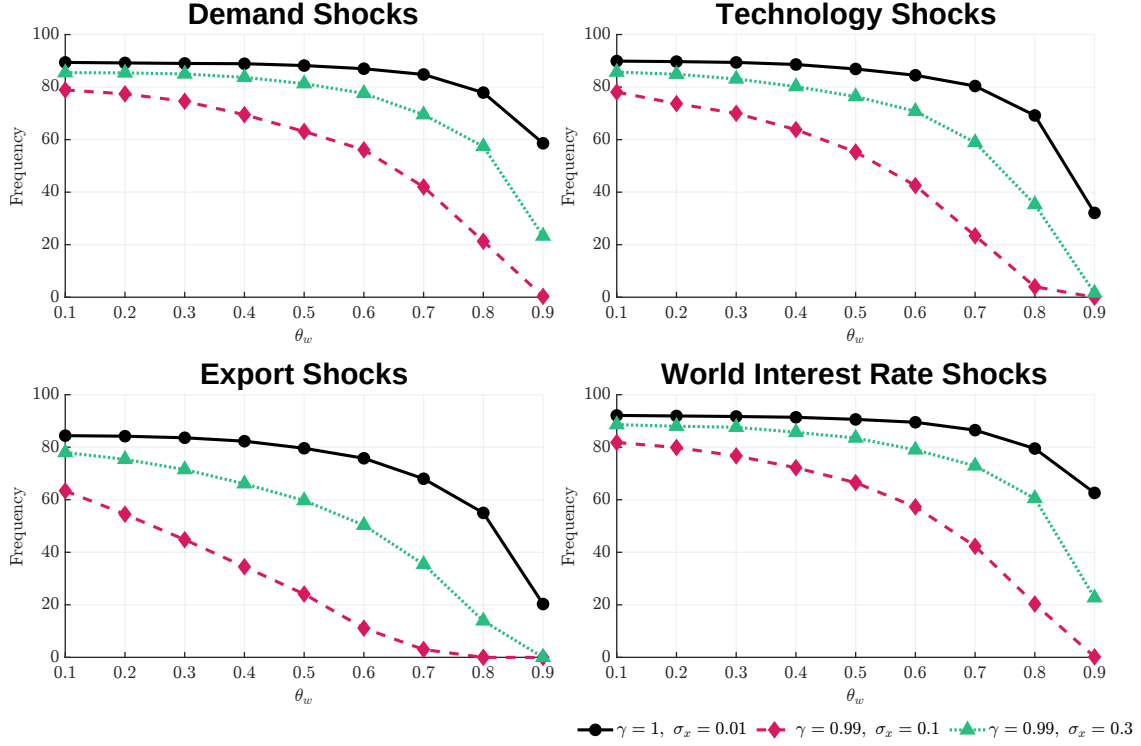


Figure B.16: Frequency of Binding DNWR Constraint with  $\gamma = 0.99$

Note: This figure plots the frequency with which the DNWR constraint binds as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) for demand (top-left), technology (top-right), export (bottom-left), and world interest rate shocks (bottom-right). The black solid line with circles corresponds to the baseline case with  $\gamma = 1$  and  $\sigma_z = 0.01$ . The red dashed line with diamonds corresponds to  $\gamma = 0.99$  and  $\sigma_z = 0.1$ , while the green dotted line with triangles corresponds to  $\gamma = 0.99$  and  $\sigma_z = 0.3$ . Higher values indicate that the DNWR constraint is binding more frequently in the simulation.

and therefore lowers the frequency with which the DNWR constraint binds. As shown in Figure B.19, the binding frequency declines substantially when annualized steady-state wage inflation increases to 2% ( $\Pi^w = 1.005$ ), particularly for technology and export shocks at high levels of wage rigidity.

As DNWR becomes less relevant, the welfare loss function gradually reverts toward the hump-shaped pattern found in Galí and Monacelli (2016), as shown in Figure B.18. To restore the relevance of DNWR in a higher-inflation environment, we increase the size of shocks to  $\sigma_z = 0.3$ . Under this calibration, the DNWR constraint binds more frequently and the monotonic relationship between welfare loss and wage rigidity is restored over most of the parameter space.

These results reinforce the state-dependent nature of DNWR: its welfare implications depend primarily on the frequency with which the constraint binds, which is jointly determined by the inflation environment, shock size, and wage rigidity.

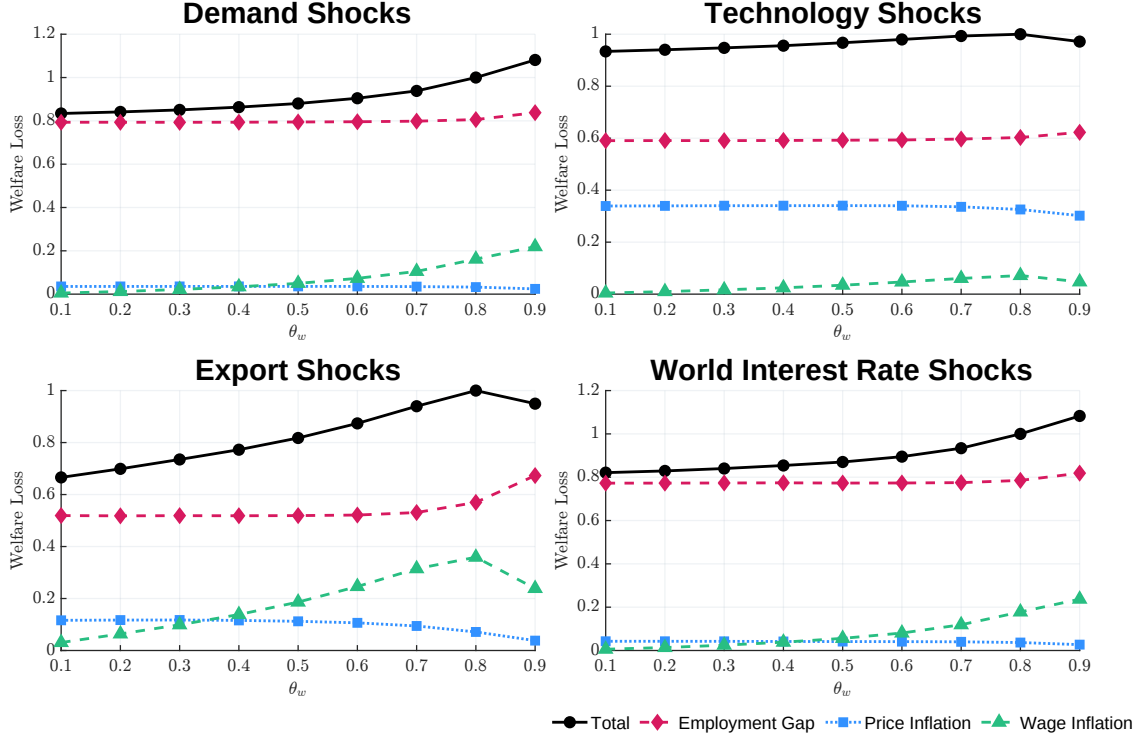


Figure B.17: Welfare Loss Decomposition with  $\gamma = 0.99$

Note: This figure decomposes welfare losses under DNWR with  $\gamma = 0.99$  as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) for demand (top-left), technology (top-right), export (bottom-left), and world interest rate shocks (bottom-right). The black solid line with circles corresponds to total welfare loss with  $\gamma = 0.99$  and  $\sigma_z = 0.3$ . Each line represents the welfare loss attributed to the employment gap (red dashed line with diamonds), price inflation (blue dotted line with squares), and wage inflation (green dashed line with triangles). Each component is normalized by baseline total welfare loss so that the sum of the components equals total welfare loss in each panel.

Figure B.20 reports welfare losses with different sizes of shock. Larger shocks increase the likelihood that the DNWR constraint binds, making DNWR more relevant for welfare outcomes.

Figure B.21 reports welfare losses following one-time positive and negative shocks, computed using the welfare loss function in Equation (37). The shock size is set to  $\sigma_z = 0.01$ .

Figures B.22, B.23, and B.24 plot welfare losses under alternative values of the labor supply elasticity, the elasticity of substitution across labor types, and the degree of openness, respectively.

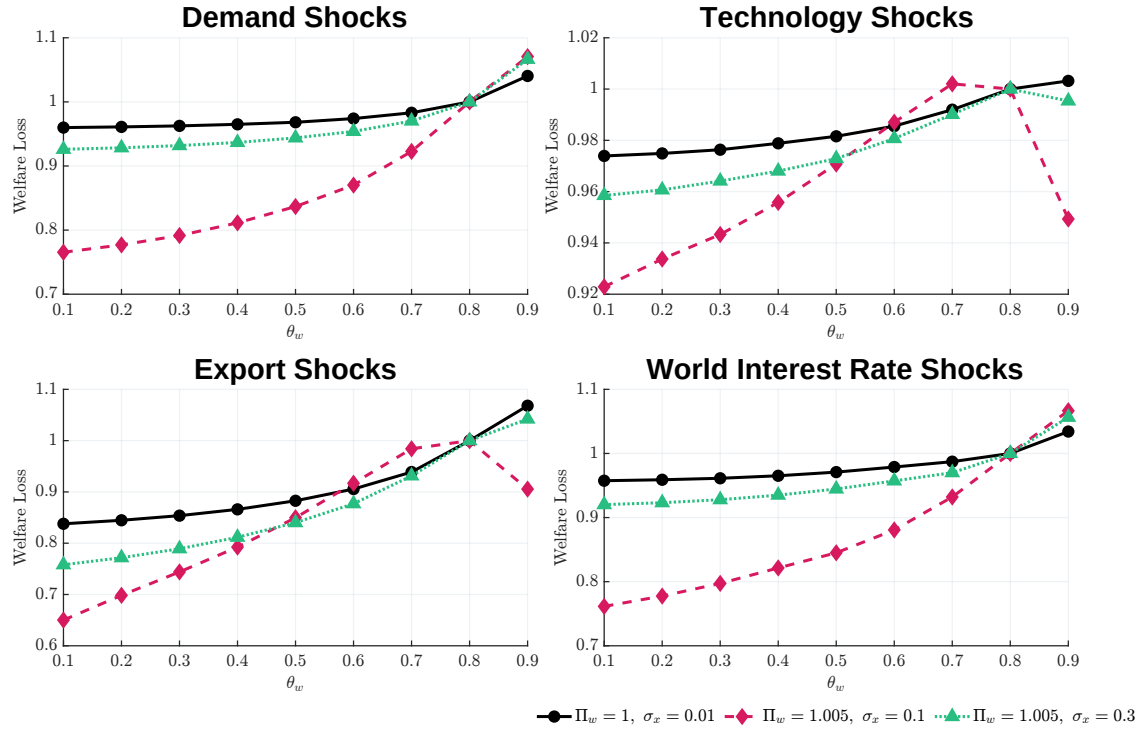


Figure B.18: Welfare Loss under Alternative Steady-State Wage Inflation Rates

Note: This figure plots welfare losses in the model with DNWR as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) for demand (top-left), technology (top-right), export (bottom-left), and world interest rate shocks (bottom-right). The black solid line with circles corresponds to the baseline calibration with zero steady-state wage inflation ( $\Pi^w = 1$ ) and shock size  $\sigma_z = 0.01$ . The red dashed line with diamonds corresponds to a specification with annualized steady-state wage inflation of 2% ( $\Pi^w = 1.005$ ) and  $\sigma_z = 0.1$ . The green dotted line with triangles corresponds to the same inflation environment ( $\Pi^w = 1.005$ ) but with a larger shock size,  $\sigma_z = 0.3$ . Welfare losses are normalized by their value at the baseline calibration,  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), with all other parameters held at their baseline values.

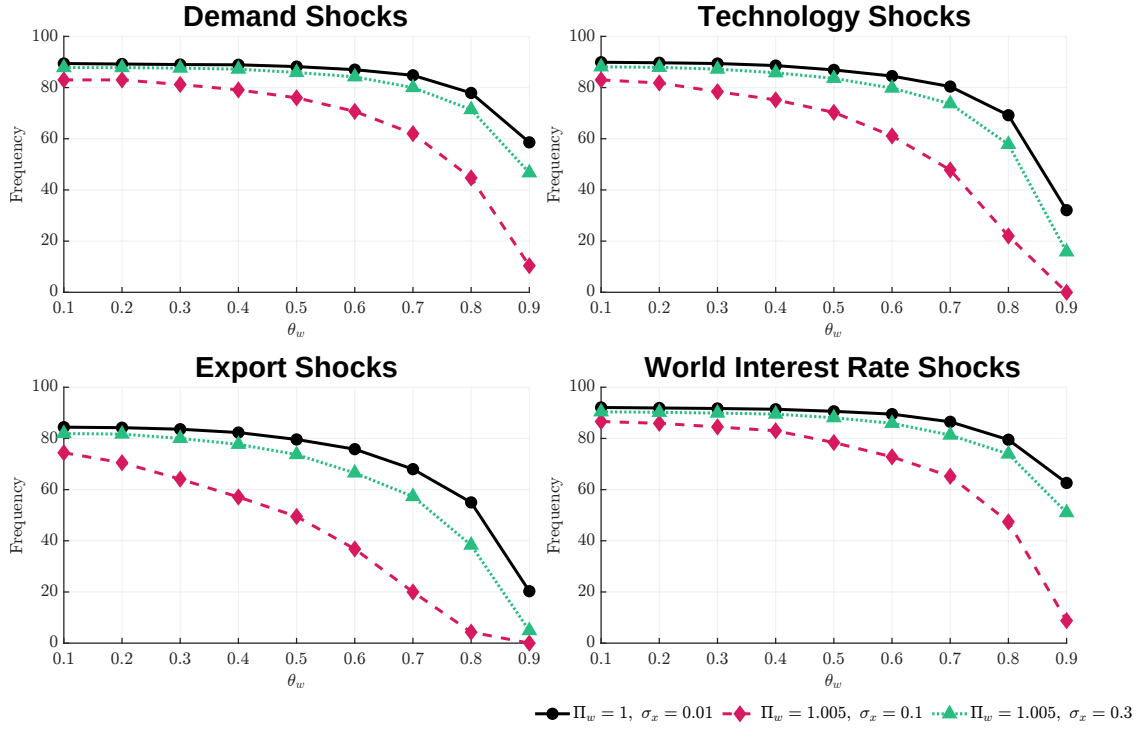


Figure B.19: Frequency of DNWR Constraint Binding under Alternative Steady-State Wage Inflation Rates

Note: This figure plots the frequency with which the DNWR constraint binds as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) for demand (top-left), technology (top-right), export (bottom-left), and world interest rate shocks (bottom-right). The black solid line with circles corresponds to the baseline calibration with zero steady-state wage inflation ( $\Pi^w = 1$ ) and shock size  $\sigma_z = 0.01$ . The red dashed line with diamonds corresponds to a specification with annualized steady-state wage inflation of 2% ( $\Pi^w = 1.005$ ) and  $\sigma_z = 0.1$ . The green dotted line with triangles corresponds to the same inflation environment ( $\Pi^w = 1.005$ ) but with a larger shock size,  $\sigma_z = 0.3$ . Higher steady-state inflation reduces the frequency with which the DNWR constraint binds by lowering the likelihood that nominal wage cuts are required.

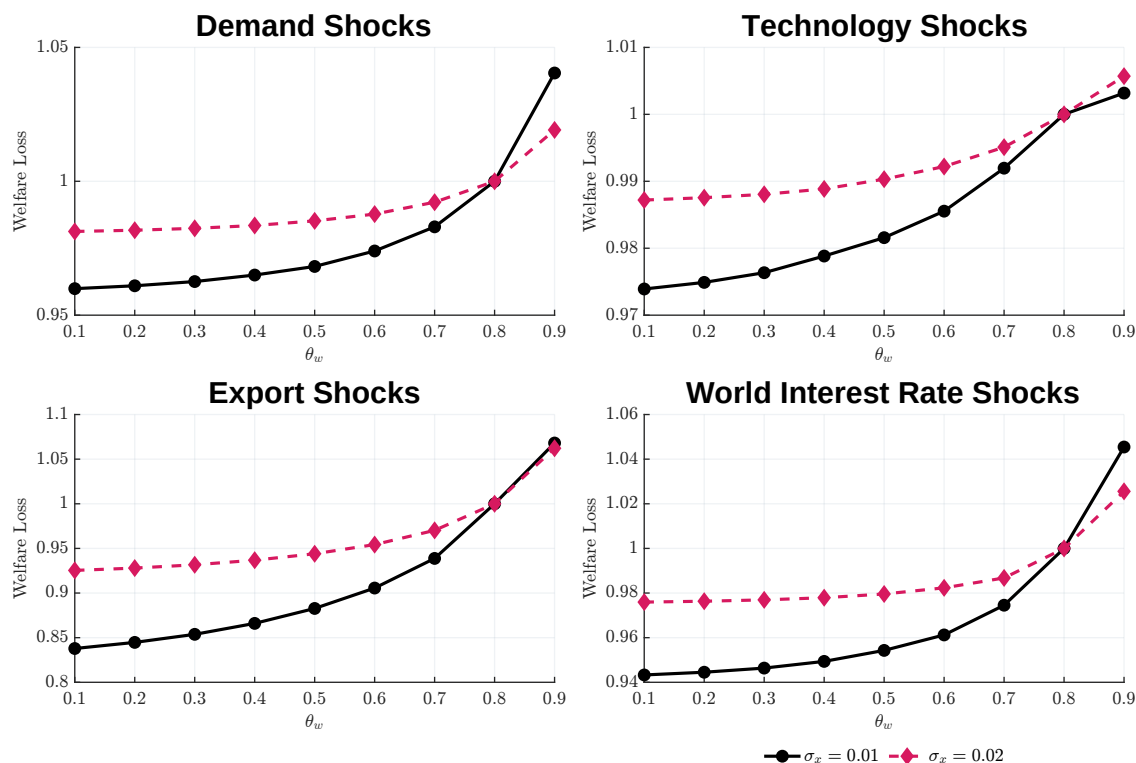


Figure B.20: Welfare Loss under Alternative Shock Sizes with DNWR

Note: This figure plots welfare losses in the model with DNWR as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) for demand (top-left), technology (top-right), export (bottom-left), and world interest rate shocks (bottom-right). The blue solid line with circles corresponds to the baseline calibration with shock size  $\sigma_z = 0.01$ , while the red dashed line with diamonds corresponds to  $\sigma_z = 0.02$ . Welfare losses are normalized by their value at the baseline calibration,  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), with all other parameters held at their baseline values.

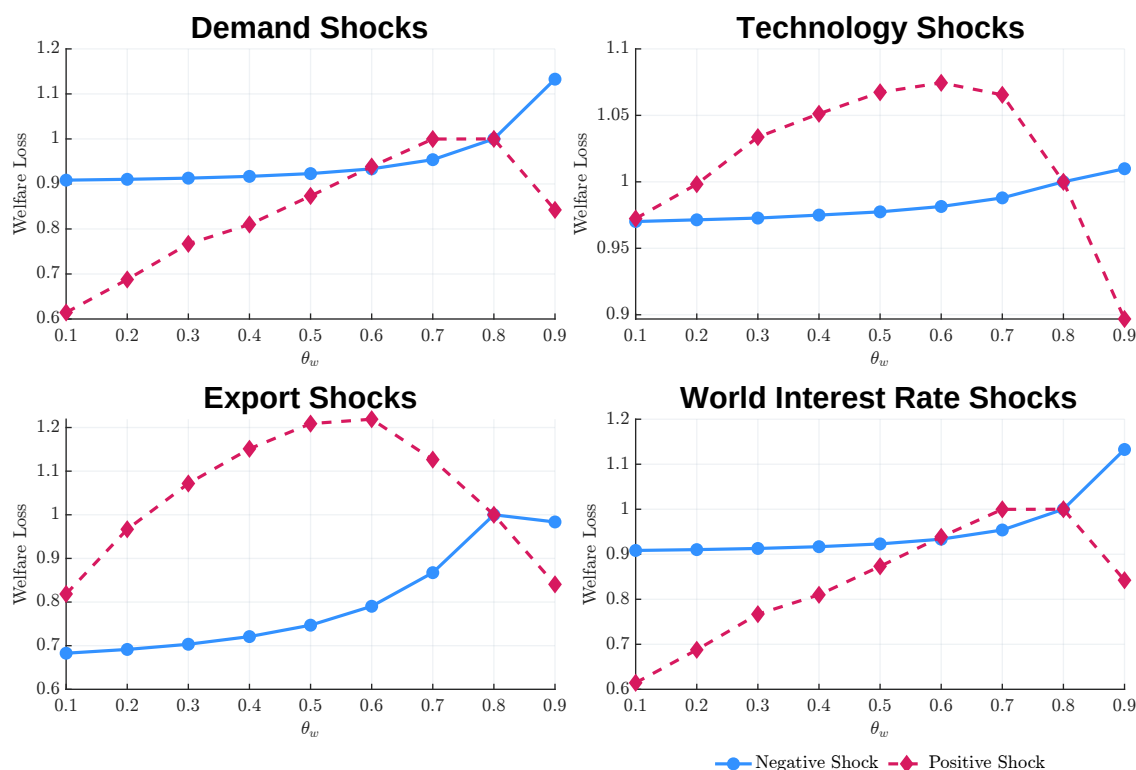


Figure B.21: Welfare Loss under Positive and Negative Shocks with DNWR

Note: This figure plots welfare losses as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) for demand (top-left), technology (top-right), export (bottom-left), and world interest rate shocks (bottom-right). The blue solid line with circles corresponds to negative shocks, while the red dashed line with diamonds corresponds to positive shocks. Here, We define positive (negative) shocks as shocks that increase (decrease) desired wage adjustment. Welfare losses are normalized by their value at the baseline calibration,  $\theta_w = 0.8$  ( $\phi_w = 864.96$ ), with all other parameters held at their baseline values.

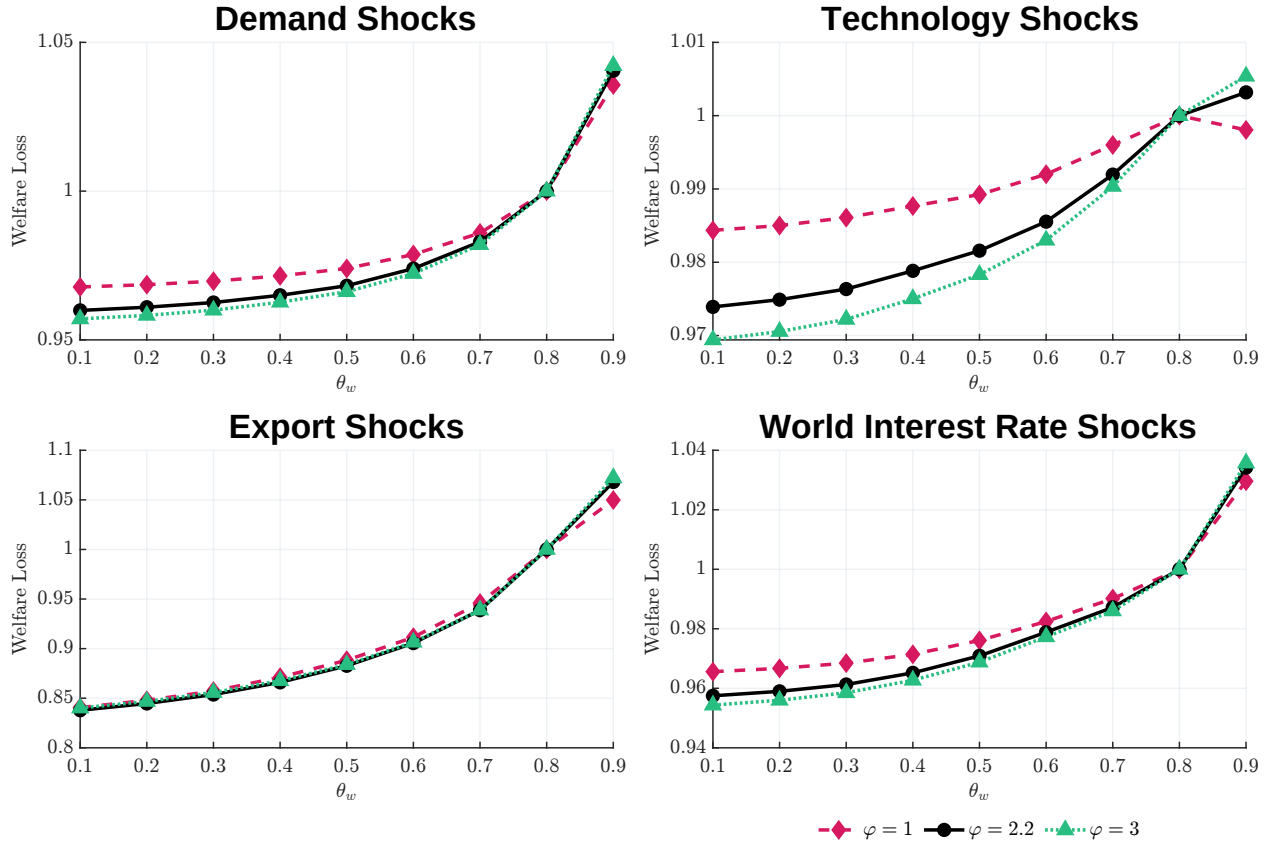


Figure B.22: Welfare Loss with Different Labor Supply Elasticity

Note: This figure plots welfare losses in the model with DNWR as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) for demand (top-left), technology (top-right), export (bottom-left), and world interest rate shocks (bottom-right). The baseline calibration sets the labor supply elasticity to  $\varphi = 2.2$ . The red dashed line with diamonds corresponds to  $\varphi = 1$ , while the green dotted line with triangles corresponds to  $\varphi = 3$ . Welfare losses are normalized by their value at the baseline calibration,  $\theta_w = 0.8$ , with all other parameters held at their baseline values.

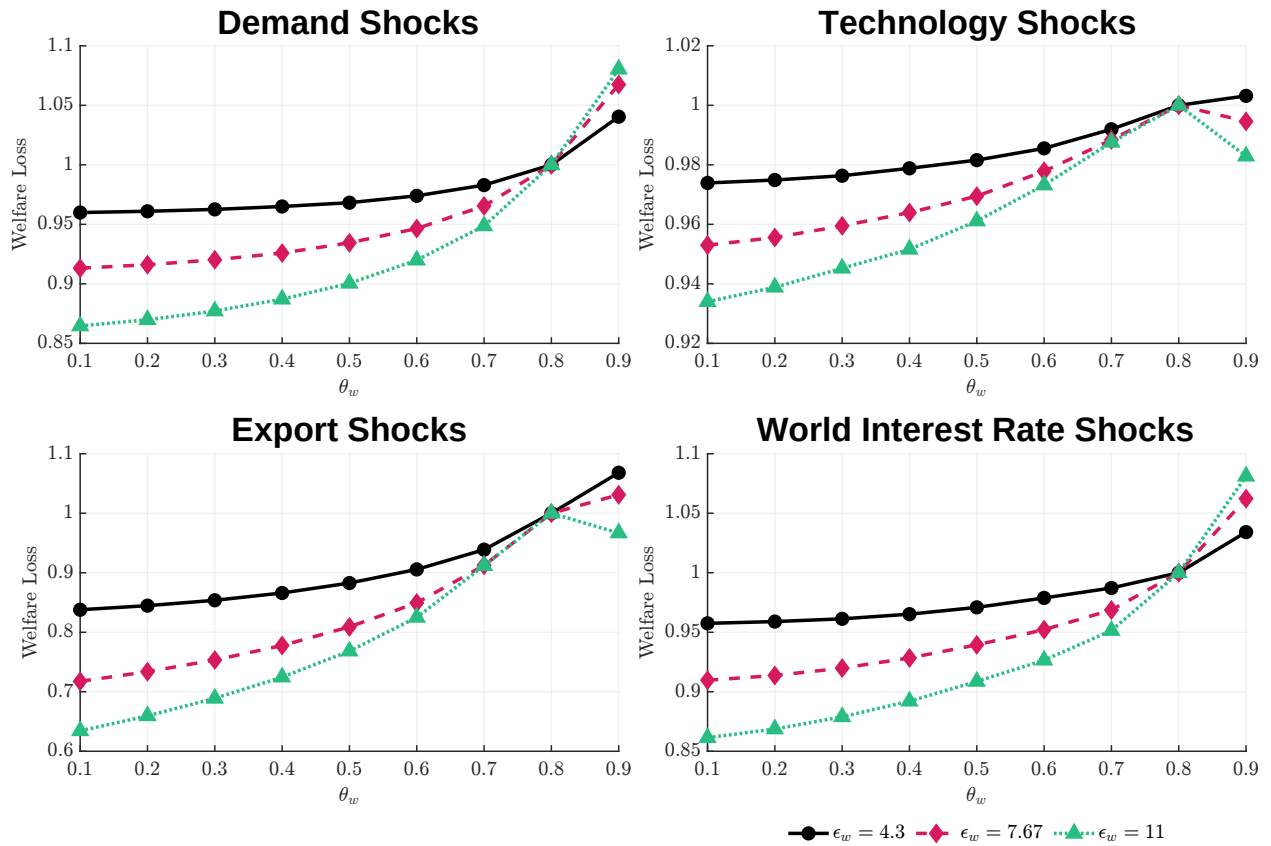


Figure B.23: Welfare Loss under Alternative Elasticities of Substitution across Labor Types

Note: This figure plots welfare losses in the model with DNWR as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) for demand (top-left), technology (top-right), export (bottom-left), and world interest rate shocks (bottom-right). The baseline calibration sets the elasticity of substitution across labor types to  $\epsilon_w = 4.3$ , implying a steady-state wage markup of 1.3. The red dashed line with diamonds corresponds to  $\epsilon_w = 7.67$ , while the green dotted line with triangles corresponds to  $\epsilon_w = 11$ . Welfare losses are normalized by their value at the baseline calibration,  $\theta_w = 0.8$ , with all other parameters held at their baseline values.

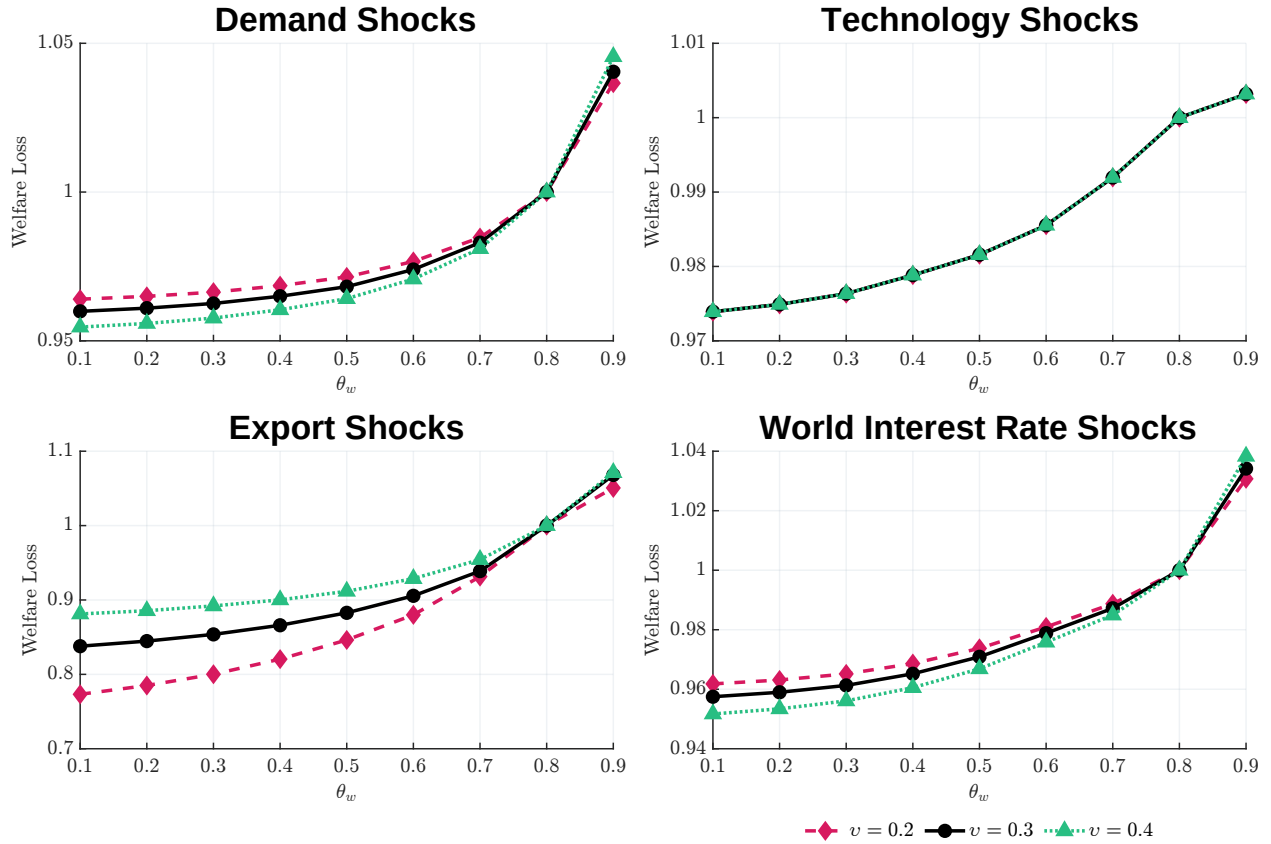


Figure B.24: Welfare Loss under Alternative Degrees of Openness

Note: This figure plots welfare losses in the model with DNWR as a function of the Calvo wage rigidity parameter ( $\theta_w$ ) for demand (top-left), technology (top-right), export (bottom-left), and world interest rate shocks (bottom-right). The baseline calibration sets the degree of openness to foreign goods to  $v = 0.3$ , implying that 30% of consumption goods are imported. The red dashed line with diamonds corresponds to  $v = 0.2$ , while the green dotted line with triangles corresponds to  $v = 0.4$ . Welfare losses are normalized by their value at the baseline calibration,  $\theta_w = 0.8$ , with all other parameters held at their baseline values.

## C Additional Results on Aggregate Demand and Supply under DNWR

Figure C.25 presents the impulse responses of employment to a negative demand shock (left panel) and a positive productivity shock (right panel) under two different monetary regimes: a currency union (second row) and strict inflation targeting (third row).

In response to a negative demand shock under a currency union regime (left panel, second row), employment declines more in the economy with DNWR. This result reflects the flatter Phillips curve under DNWR: downward rigidity in wages suppresses price inflation adjustment, causing a larger share of macroeconomic adjustment to occur through employment. By contrast, under a strict inflation-targeting regime (left panel, third row), monetary policy responds endogenously to weaker inflationary pressures by lowering interest rates, thereby supporting aggregate demand and employment. The additional distortion generated by DNWR is largely offset in this case, and employment responses become quantitatively very similar across the two models.

In response to a positive technology shock, employment declines in a currency union (right panel, second row), reflecting the upward-sloping AD curve. A positive supply shock lowers inflation, raising the real interest rate when monetary policy does not respond to domestic conditions. This suppresses aggregate demand and generates a short-run contraction in employment (Cacciatore et al., 2016). The presence of DNWR modestly amplifies this effect by flattening the AD curve, although the quantitative impact remains limited.

Under a strict inflation-targeting regime (right panel, third row), monetary policy responds to lower inflationary pressures by implementing expansionary monetary policy to stabilize inflation. As a result, employment increases following a positive productivity shock. The positive employment response is somewhat larger in the presence of DNWR, since DNWR limits downward wage adjustment and dampens the decline in wages during the transition dynamics.

The slopes of the AD and AS curves are influenced by both the degree of DNWR and the size of shocks. Figure C.26 plots the slopes of the AD and AS curves, defined as the regression coefficients annualized CPI inflation on employment, under different values of DNWR ( $\gamma = 0, 0.99$ , and  $1$ ) and different shock sizes ( $\sigma = 0.01, 0.1, 0.2$ , and  $0.3$ ). A higher degree of DNWR or larger shocks leads to a more pronounced flattening of the AS curve.

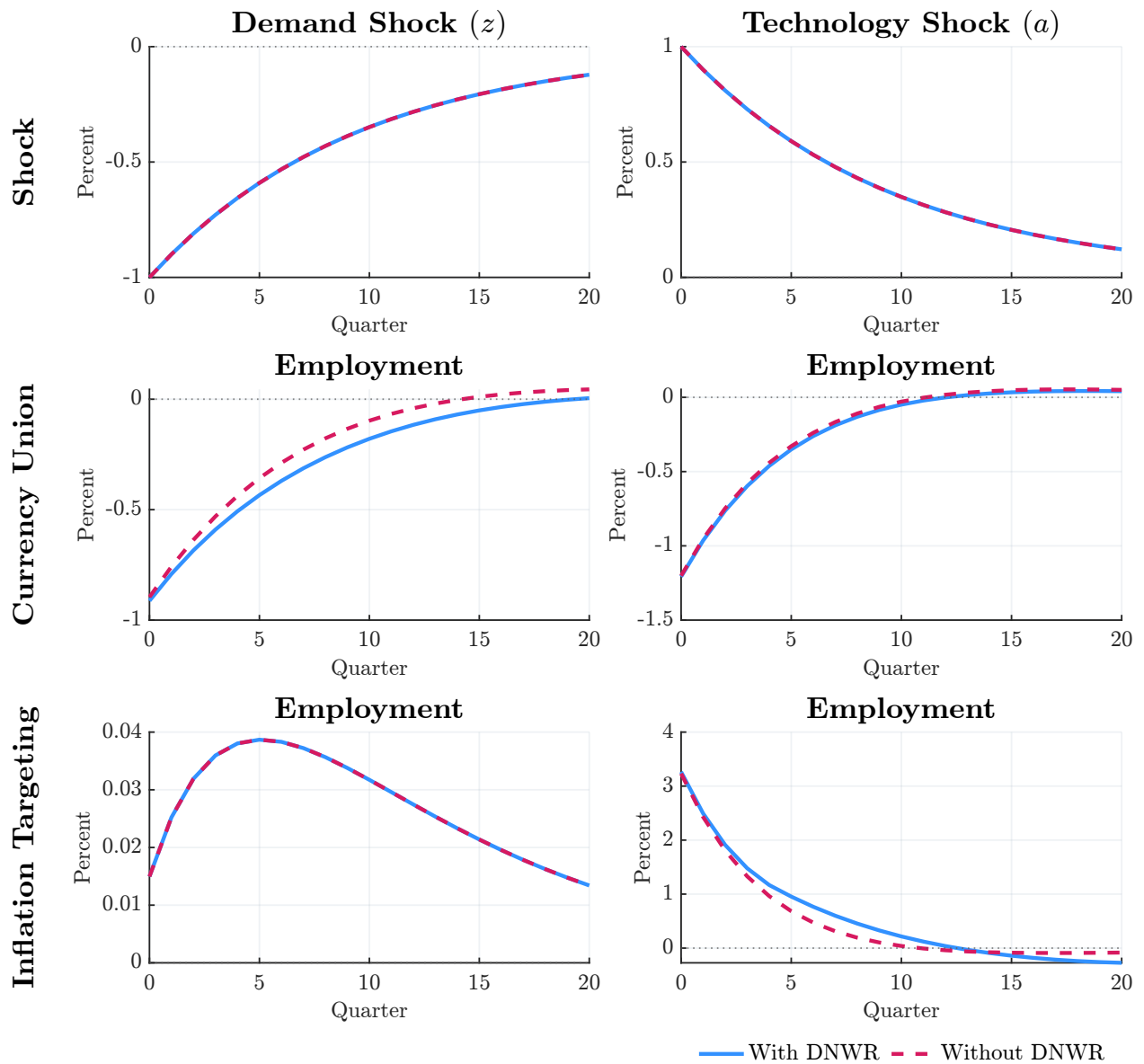
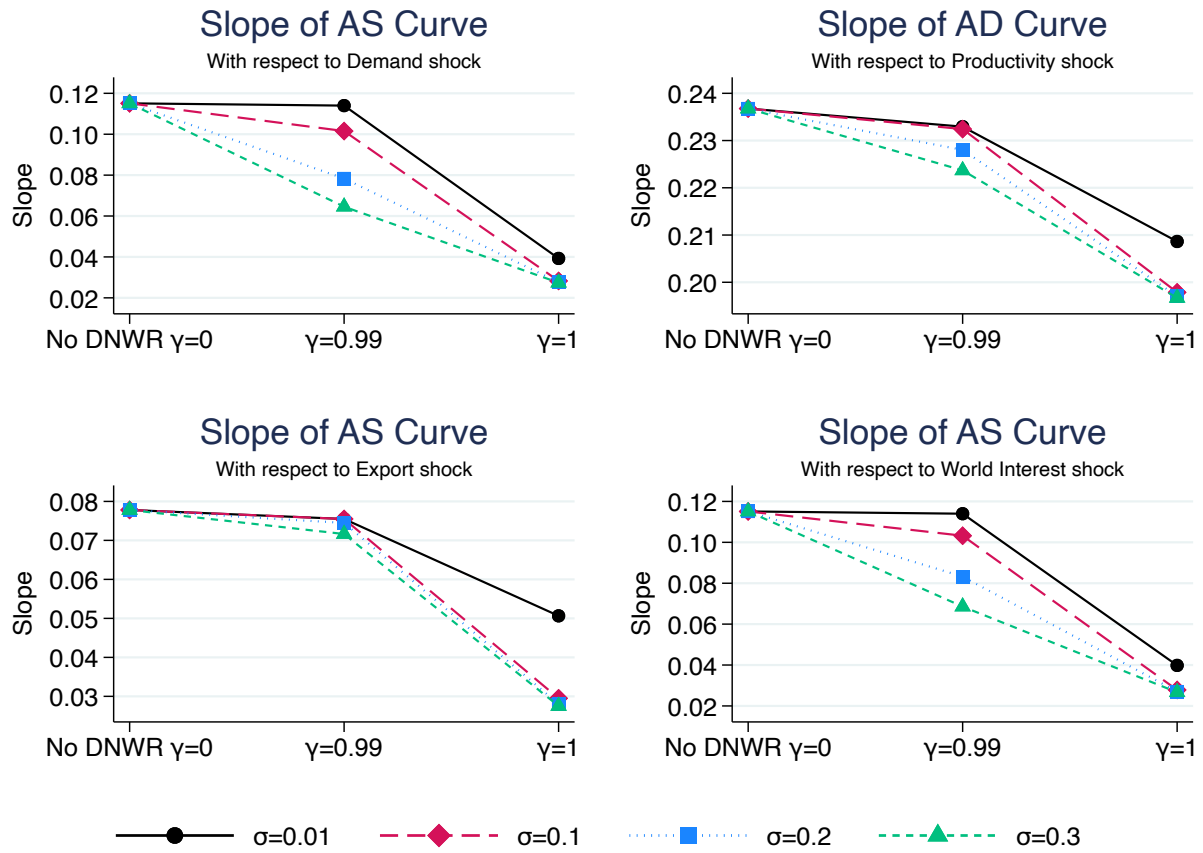


Figure C.25: Impulse Response Functions to Negative Demand and Positive Supply Shocks across Different Monetary Policy Regimes

This figure plots impulse response functions to a negative demand shock (left panel) and a positive supply shock (right panel) under different monetary policy regimes. The top panel displays the shock processes for demand (left) and supply (right). The middle panel represents the currency union regime, while the bottom panel shows the inflation-targeting regime.

Figure C.26: Slope of AD–AS Curves with Varying DNWR



Notes: The figure plots the slope of AD–AS curves as a function of DNWR ( $\gamma$ ) and shock size ( $\sigma$ ). Slopes are estimated from regressions of annualized CPI inflation rate on employment using simulated data (10,000 observations per shock). The panels correspond to demand, productivity, export, and world interest rate shocks; the AS curve is shown for all shocks except productivity, for which the AD curve is reported.

## D Linearized Equilibrium Dynamics

### D.1 Aggregate Demand Block

$$y_t = (1 - v) c_t + v (2 - v) s_t + v z_{1,t}^* \quad (\text{D.2})$$

$$c_t = (1 - v) s_t + z_t - z_{2,t}^* \quad (\text{D.3})$$

$$c_t = E_t c_{t+1} - (i_t - E_t \pi_{t+1}^p) + (1 - \rho_z) z_t \quad (\text{D.4})$$

$$s_t \equiv e_t - p_{H,t} \quad (\text{D.5})$$

$$n_t = \frac{1}{1 - \alpha} (y_t - a_t) \quad (\text{D.6})$$

### D.2 Aggregate Supply Block

$$\pi_{H,t}^p = \beta E_t \pi_{H,t+1}^p + \frac{\lambda_p \alpha}{1 - \alpha} \tilde{y}_t + \lambda_p \tilde{\omega}_t + \lambda_p v \tilde{s}_t \quad (\text{D.7})$$

where  $\lambda_p \equiv \frac{\epsilon_p Y}{\phi_p} = \frac{(1 - \theta_p)(1 - \theta_p \beta)(1 - \alpha)}{\theta_p(1 - \alpha + \alpha \epsilon_p)}$

$$\pi_{H,t}^p \equiv p_{H,t} - p_{H,t-1} \quad (\text{D.8})$$

$$\pi_t^p \equiv p_t - p_{t-1} \quad (\text{D.9})$$

$$p_t = p_{H,t} + v s_t \quad (\text{D.10})$$

$$\pi_t^w - \lambda_w \delta_t = \beta E_t (\pi_{t+1}^w - \lambda_w \gamma \delta_{t+1}) + \frac{\lambda_w \varphi}{1 - \alpha} \tilde{y}_t + \lambda_w \tilde{c}_t - \lambda_w \tilde{\omega}_t \quad (\text{D.11})$$

where  $\lambda_w \equiv \frac{\epsilon_w N}{\phi_w} = \frac{(1 - \theta_w)(1 - \beta \theta_w)}{\theta_w(1 + \epsilon_w \varphi)}$  and  $\delta_t \equiv \frac{1}{\epsilon_w N} (\Delta_t - \Delta)$

$$\delta_t (\pi_t^w - \log \gamma + \pi^w) = 0, \quad \delta_t \geq 0, \quad \pi_t^w - \log \gamma + \pi^w \geq 0 \quad (\text{D.12})$$

$$\pi_t^w \equiv w_t - w_{t-1} \quad (\text{D.13})$$

$$\omega_t \equiv w_t - p_t \quad (\text{D.14})$$

### D.3 Gaps

Variables with tilde denote deviations from their natural (i.e., flexible price and wage) equilibrium counterparts:

$$\tilde{y}_t \equiv y_t - y_t^n \quad (\text{D.15})$$

$$\tilde{s}_t \equiv s_t - s_t^n \quad (\text{D.16})$$

$$\tilde{c}_t \equiv c_t - c_t^n \quad (\text{D.17})$$

$$\tilde{\omega}_t \equiv \omega_t - \omega_t^n \quad (\text{D.18})$$

$$\tilde{n}_t \equiv n_t - n_t^n \quad (\text{D.19})$$

### D.4 Natural Variables

$$n_t^n = \frac{v}{1 + \varphi} (z_{1,t}^* + z_{2,t}^* - z_t) \quad (\text{D.20})$$

$$y_t^n = a_t + (1 - \alpha) n_t^n \quad (\text{D.21})$$

$$s_t^n = a_t - z_t + z_{2,t}^* - (\alpha + \varphi) n_t^n \quad (\text{D.22})$$

$$c_t^n = z_t + (1 - v) s_t^n - z_{2,t}^* \quad (\text{D.23})$$

$$\omega_t^n = a_t - \alpha n_t^n - v s_t^n \quad (\text{D.24})$$

### D.5 World Interest Rate

$$i_t^* = (1 - \rho_2^*) z_{2,t}^* \quad (\text{D.25})$$

### D.6 Exogenous Processes

$$a_t = \rho_a a_{t-1} + \sigma_a \varepsilon_t^a \quad (\text{D.26})$$

$$z_t = \rho_z z_{t-1} + \sigma_z \varepsilon_t^z \quad (\text{D.27})$$

$$z_{1,t} = \rho_{z,1} z_{1,t-1} + \sigma_{z,1} \varepsilon_t^{z,1} \quad (\text{D.28})$$

$$z_{2,t} = \rho_{z,2} z_{2,t-1} + \sigma_{z,2} \varepsilon_t^{z,2} \quad (\text{D.29})$$